KMOU Division of Marine System Engineering

Introduction to Internal Combustion Engine

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Lesson content for each week

Ch0_Introduction to the Class

Ch1_Overview of the Internal Combustion Engine

Ch2_Thermodynamics and Compressed Air System

Ch3_Marine Fuels and Internal Fuel Oil System

Ch4_Intake and Exhaust System

Ch5_Turbocharger

Midterm

Ch6_Piston Fittings and Crankshaft

Ch7_Lube Oil System

Ch8_Cooling Water System

Ch9_Introduction to the Main Engine

Ch10_Alternative Fuels and Alternative Fuel Systems

Ch11_Environmental Regulations and Exhaust Gas Aftertreatment Systems

Ch12_Tools, Apprentice Engineer Duties, and Career Direction

Final Exam

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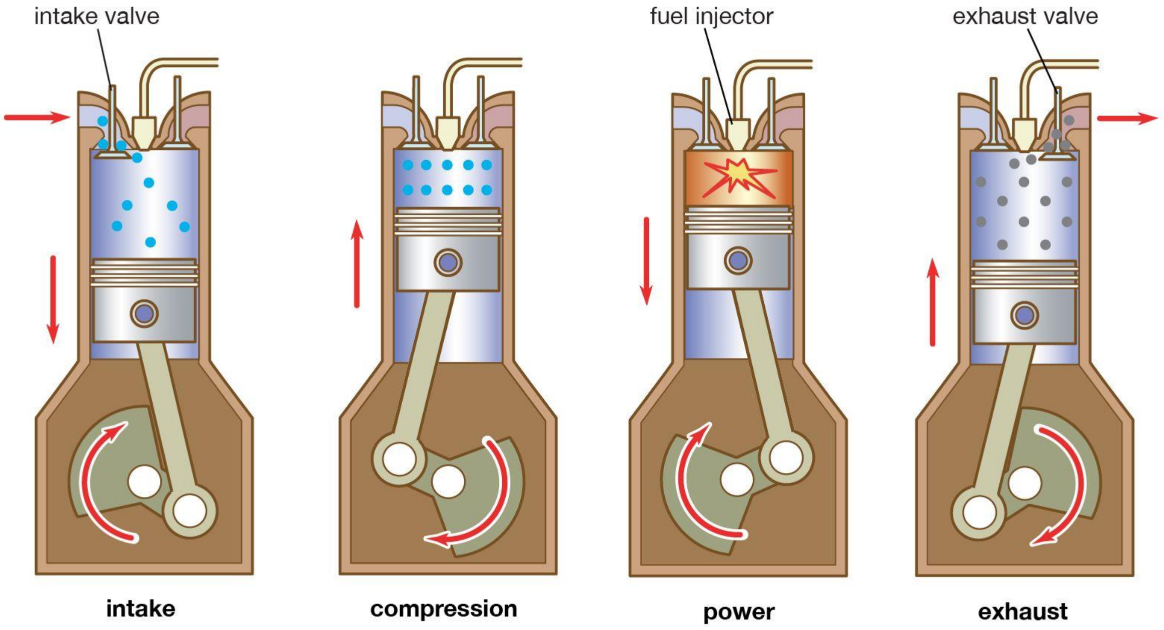
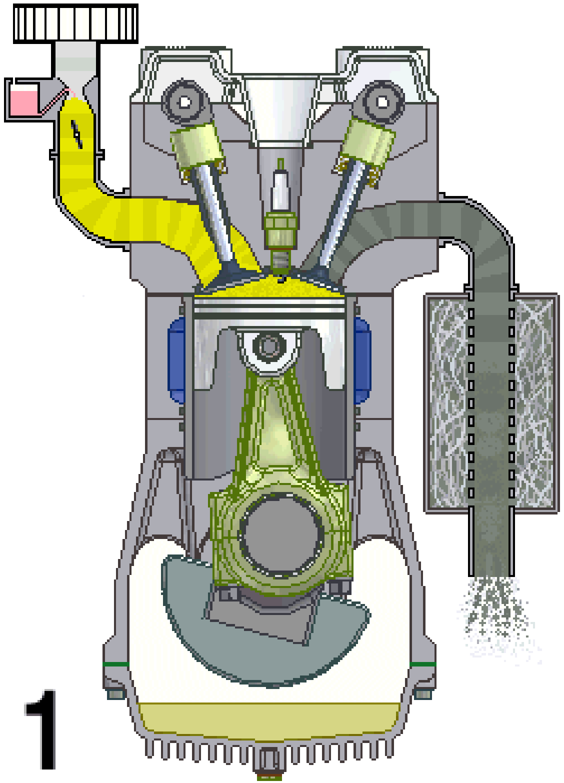
4-stroke ICE (G/E)

The engine of generator engine (G/E) on ship is a 4-stroke ICE.

What is 4-stroke ICE?





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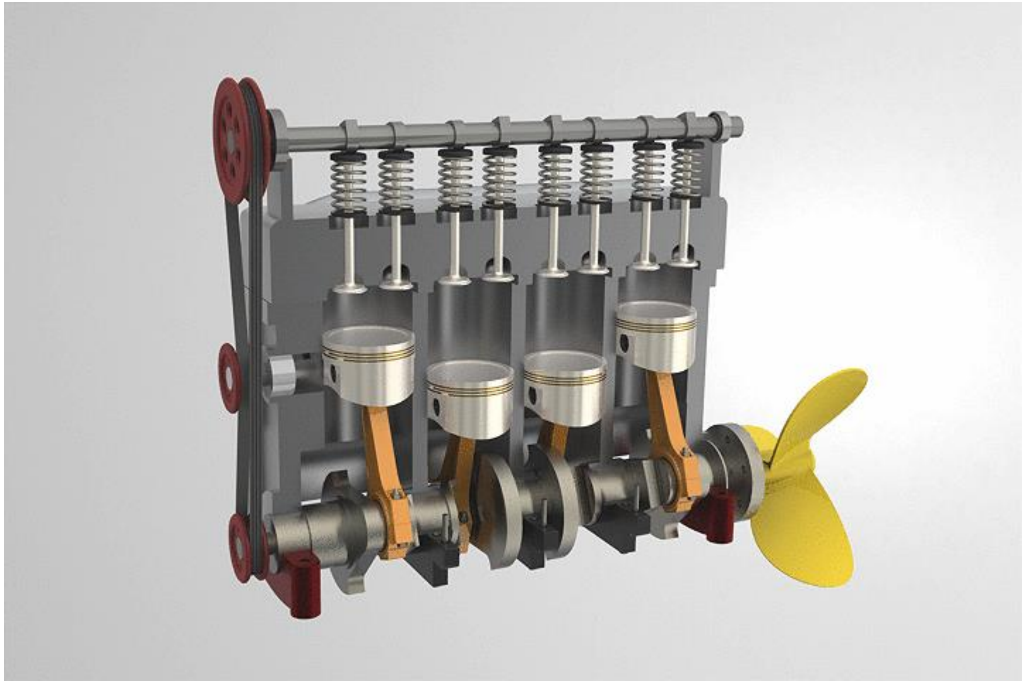
What is 4-stroke ICE?



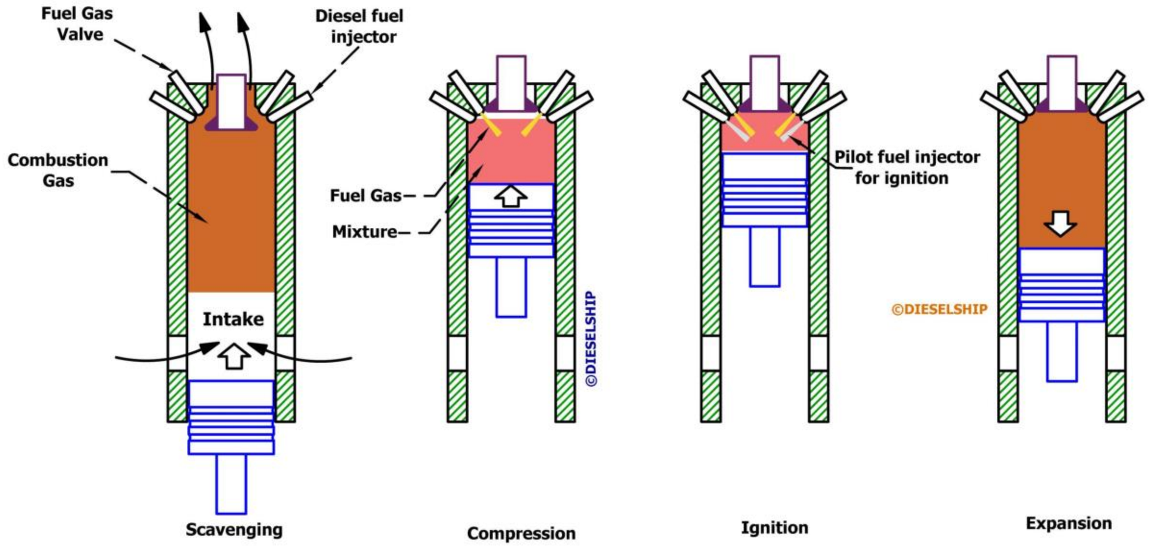
Real G/E is little bit different with left animation.

This animation is just for reference.

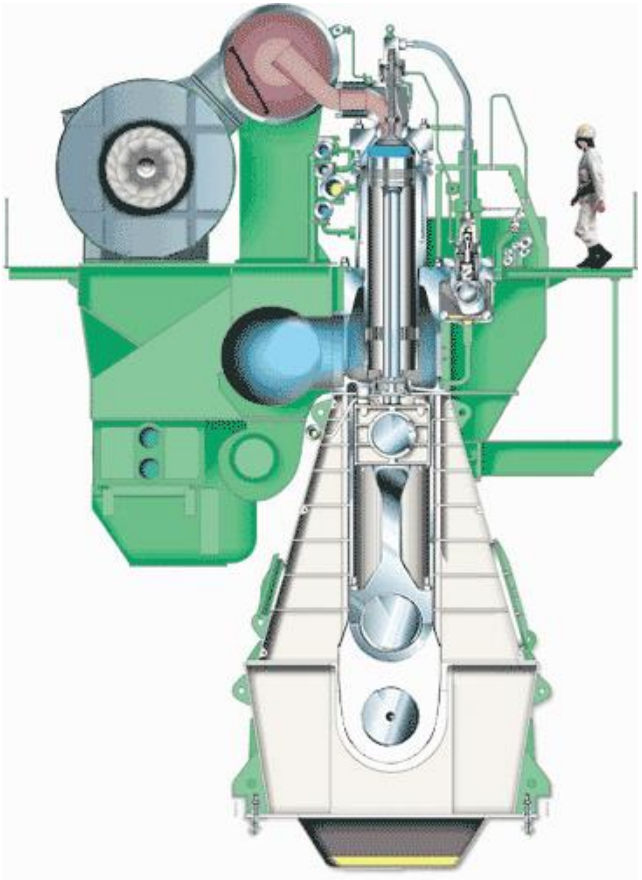
2-stroke ICE (M/E)

The main engine (M/E) on ship is a 2-stroke ICE.

What is 2-stroke ICE?



2-stroke dual fuel engine



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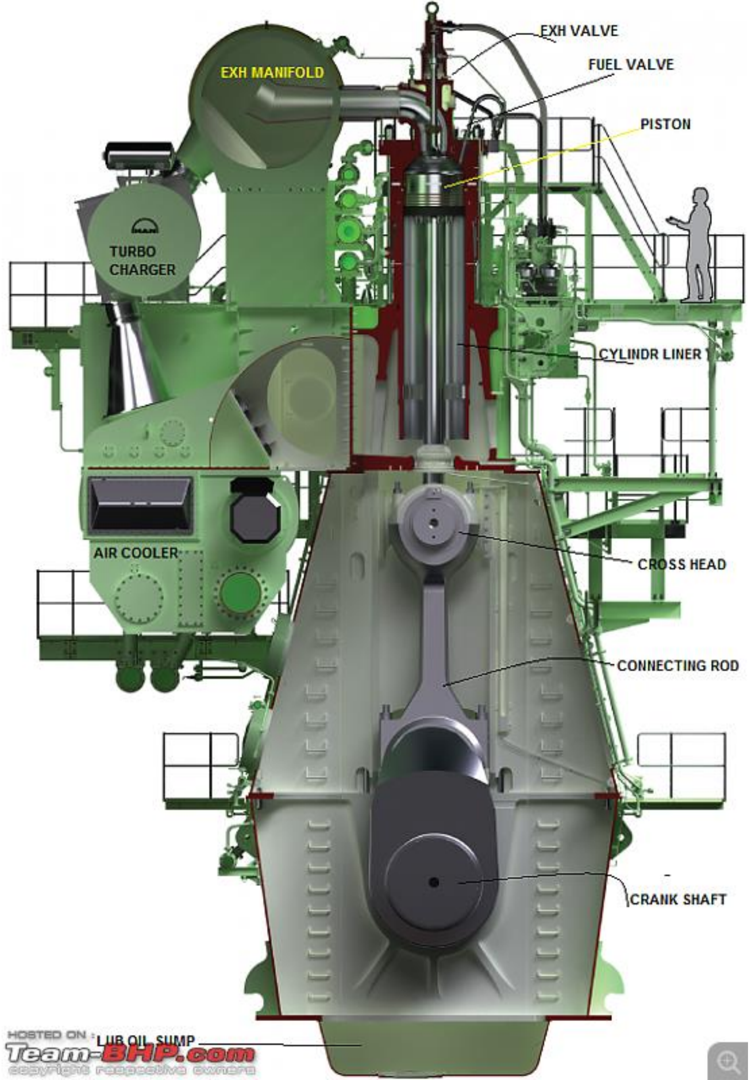
2-stroke ICE



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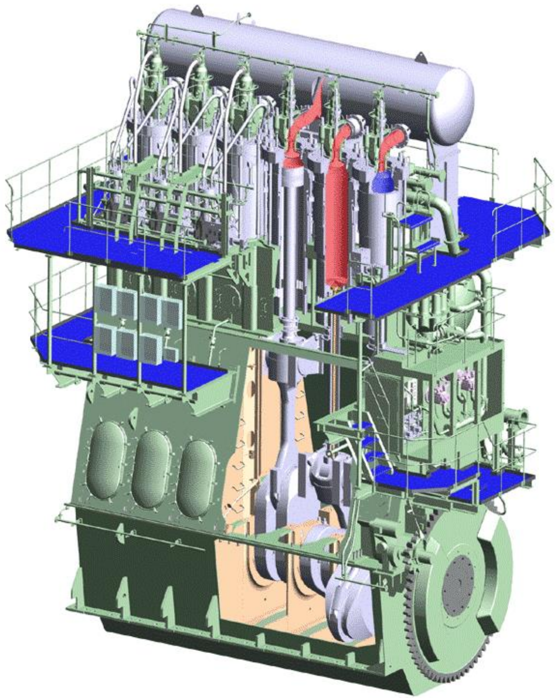
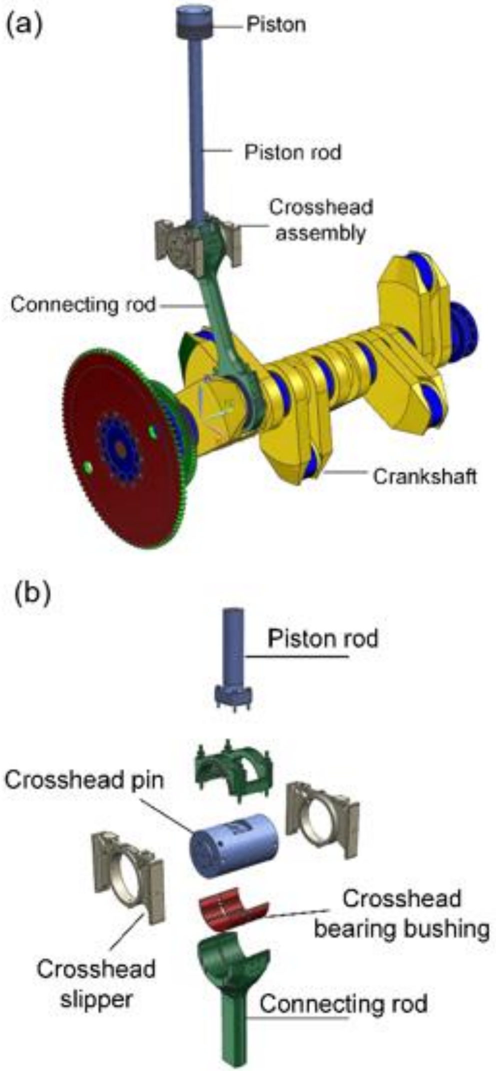
2-stroke ICE (M/E)

The main engine (M/E) on ship is a 2-stroke ICE.

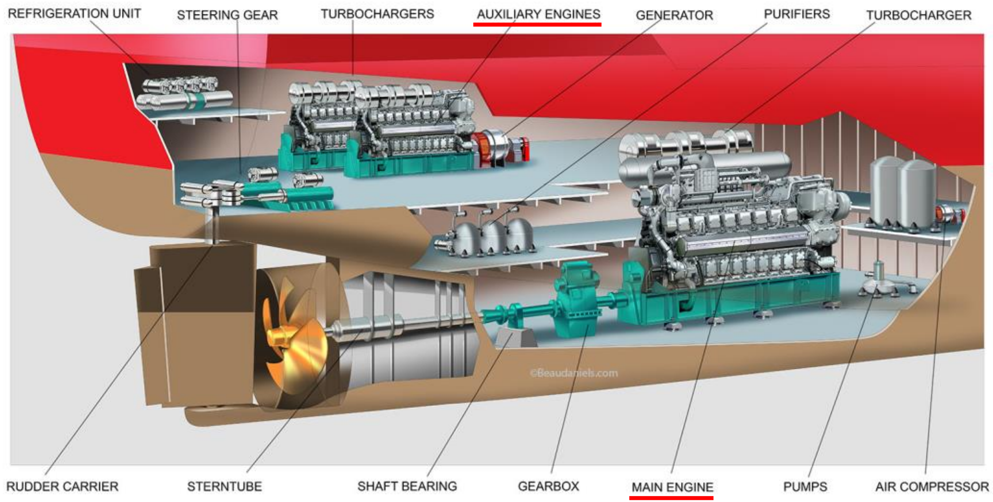
What is 2-stroke ICE?

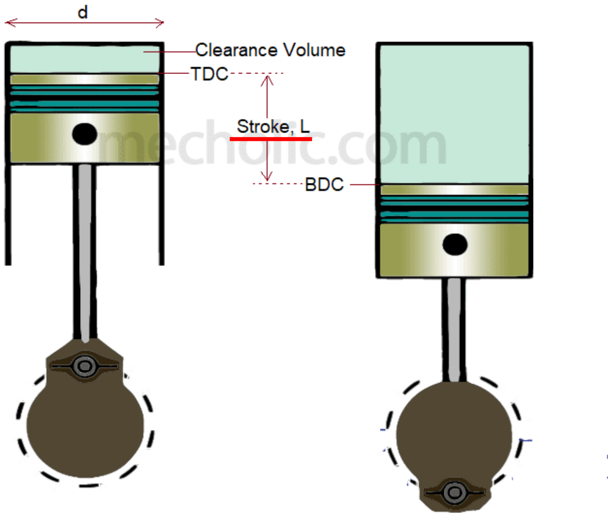




Difference between G/E and M/E

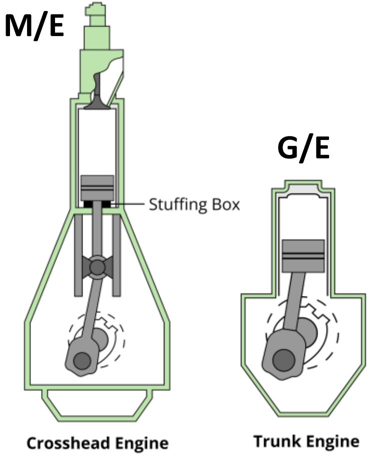
G/E	M/E
The main purpose is to produce electricity.	The main purpose is to propel ships.
4-stroke (Intake, compression, explosion, and exhaust are performed in four strokes.)	2-stroke (Scavenging and compression are performed in one stroke, ignition and expansion are performed in one stroke)

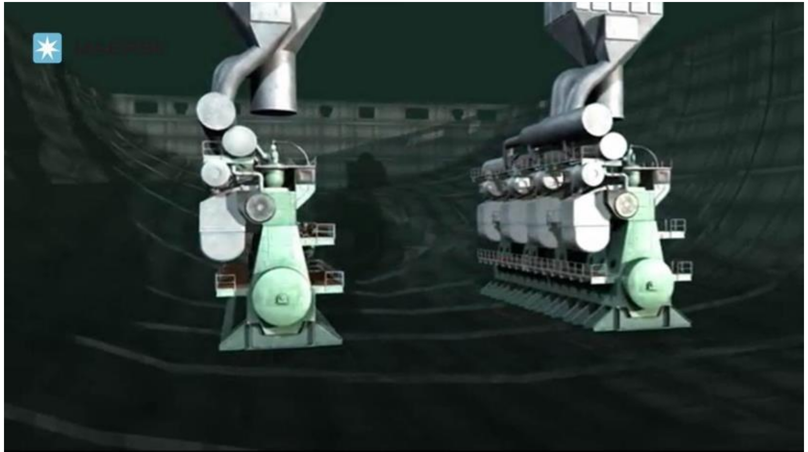




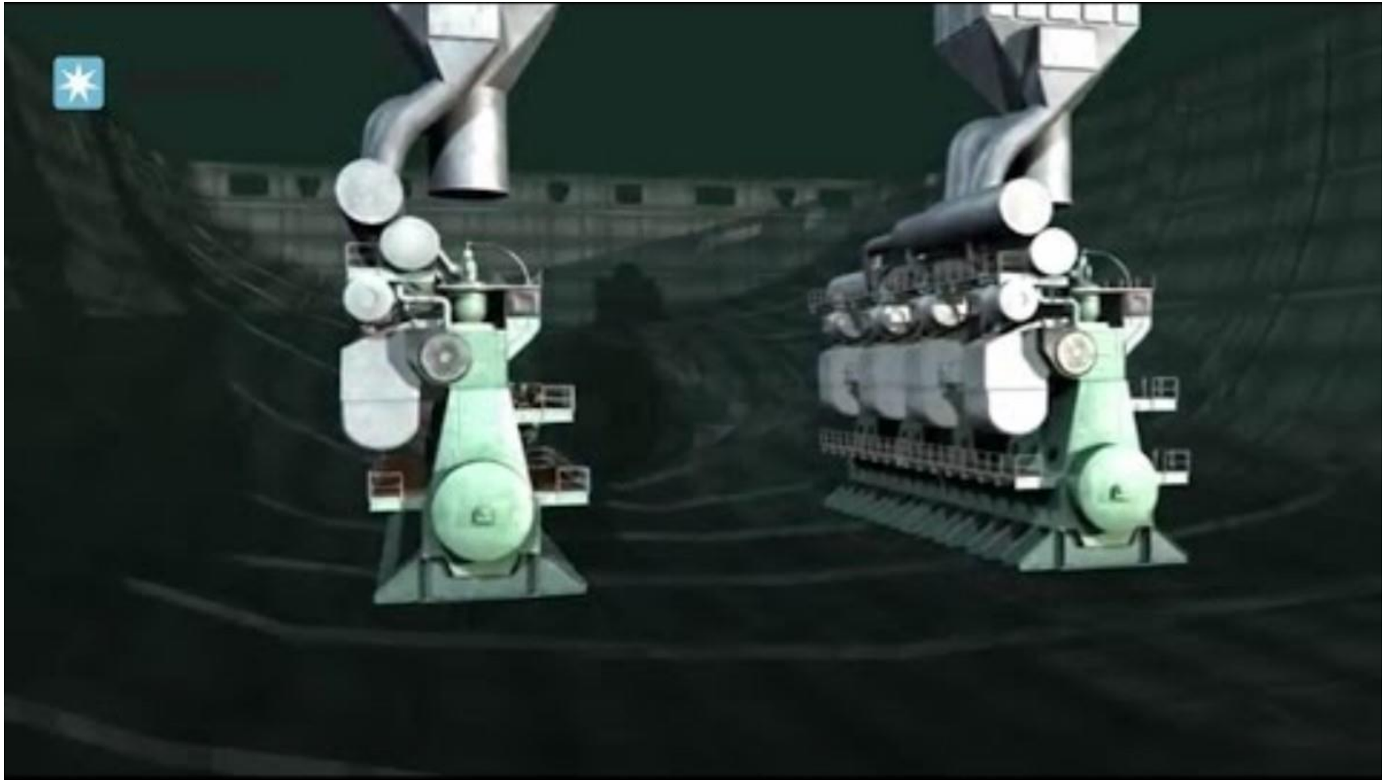
Difference between G/E and M/E

G/E	M/E
Trunk type engine	Crosshead type engine (with stuffing box)
Smaller and lighter than M/E	Larger and heavier than G/E
Typically, 3 to 4 G/Es are installed on a ship.	Typically, one M/E is installed on a ship. In the case of twin screw propulsion type, two are installed (not common case).

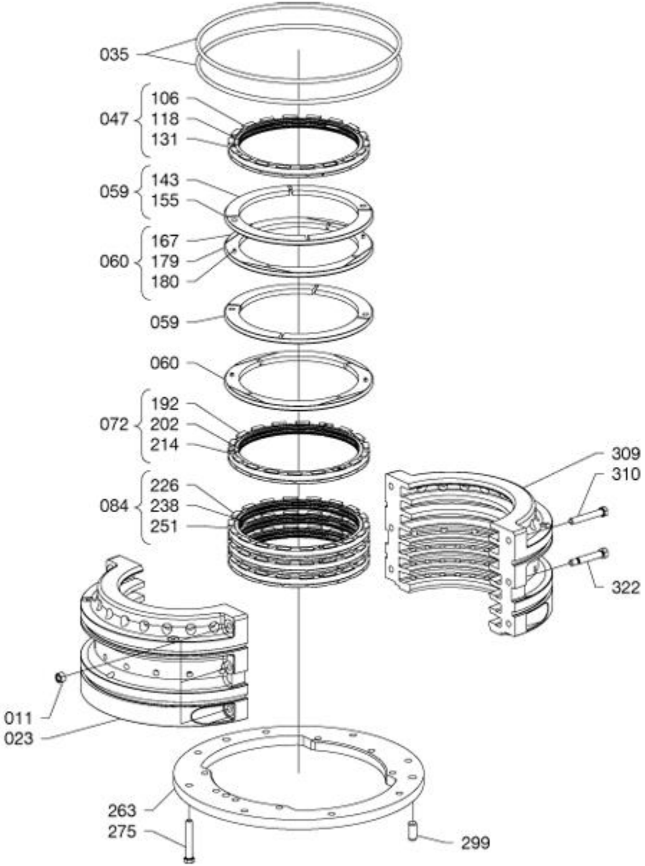




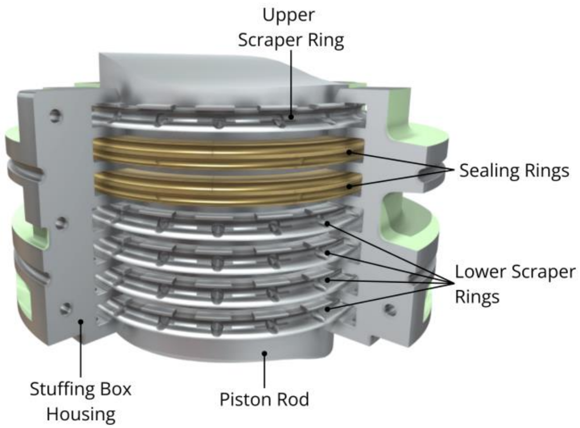
Difference between G/E and M/E



Stuffing box

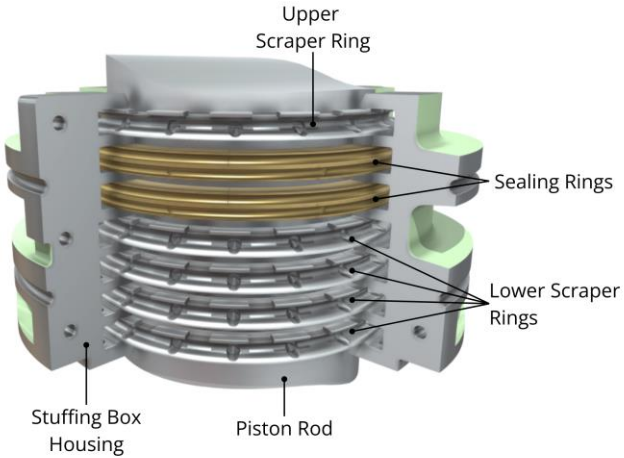


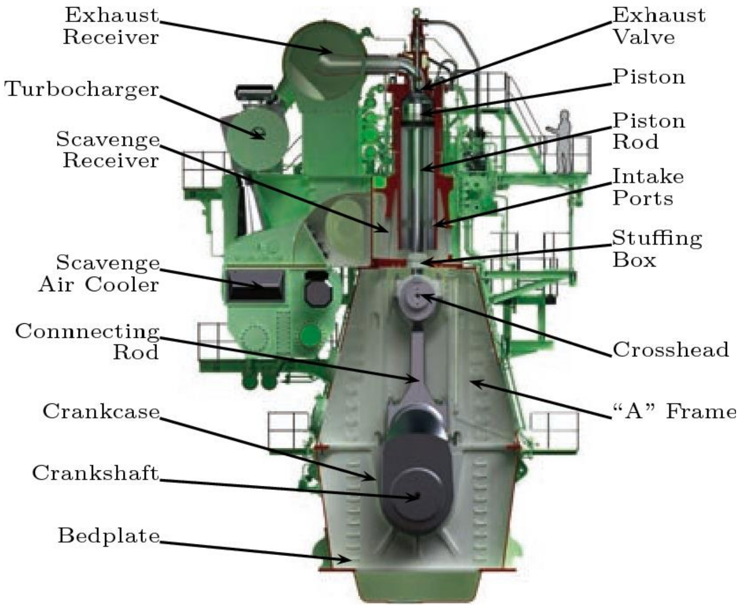
Item No.	Qty	Item Designation
011	-	Nut
023	-	Housing
035	-	O-ring
047	-	Scraper ring, complete
059	-	Sealing ring, complete
060	-	Sealing ring, complete
072	-	Scraper ring, complete
084	-	Scraper ring, complete
106	-	Base ring
118	-	Lamella
131	-	Tension spring
143	-	Segment, sealing ring
155	-	Tension spring
167	-	Segment, sealing ring
179	-	Tension spring
180	-	Spring pin
192	-	Scraper ring
202	-	Lamella
214	-	Tension spring
226	-	Base ring, scraper ring
238	-	Lamella
251	-	Tension spring
263	-	Flange
275	-	Screw
299	-	Guide pin
309	-	Housing
310	-	Screw
322	-	Fitted bolt

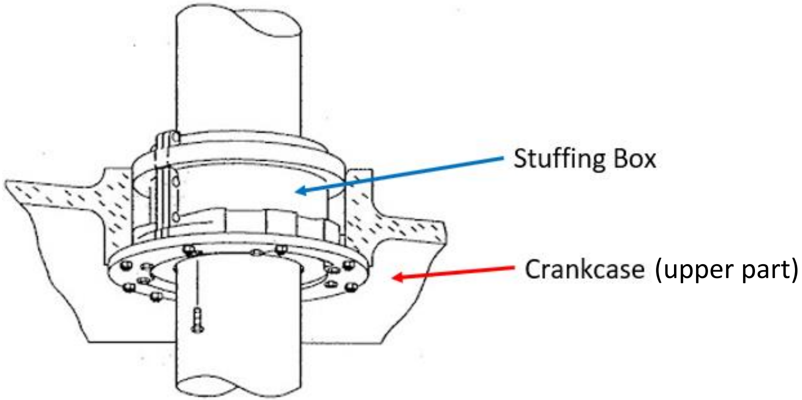


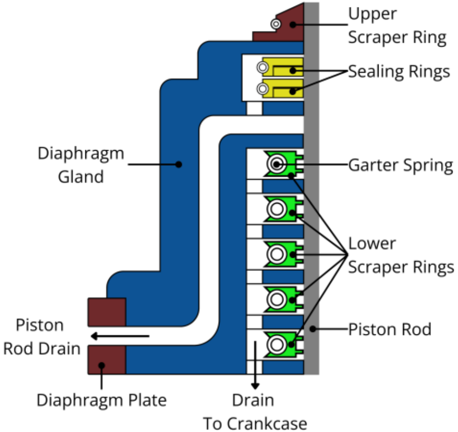
The stuffing box is a critical sealing assembly that prevents the transfer of oil, debris, and combustion gases between the engine's cylinder/scavenge space and the crankcase, and vice versa.

Stuffing box









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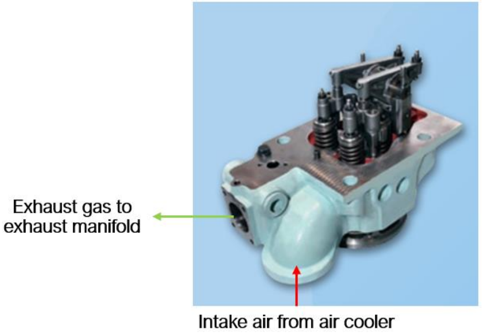
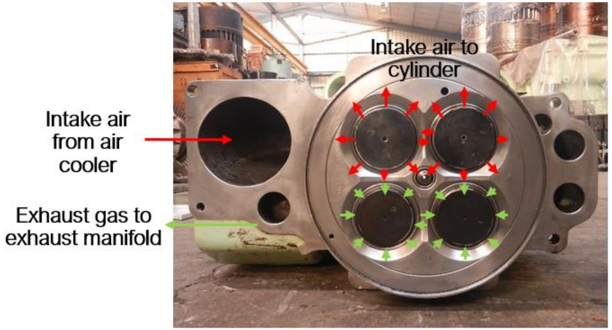
Stuffing box

Stuffing box

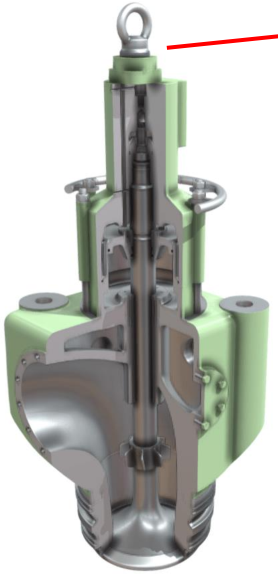
Difference between G/E and M/E

G/E	M/E
Two intake valves and two exhaust valves are installed in the cylinder head.	One exhaust valve is installed on the cylinder cover.

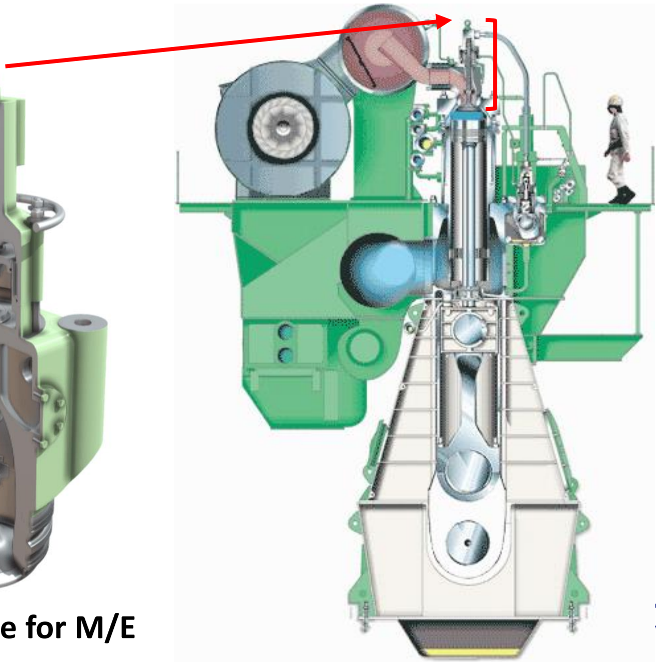




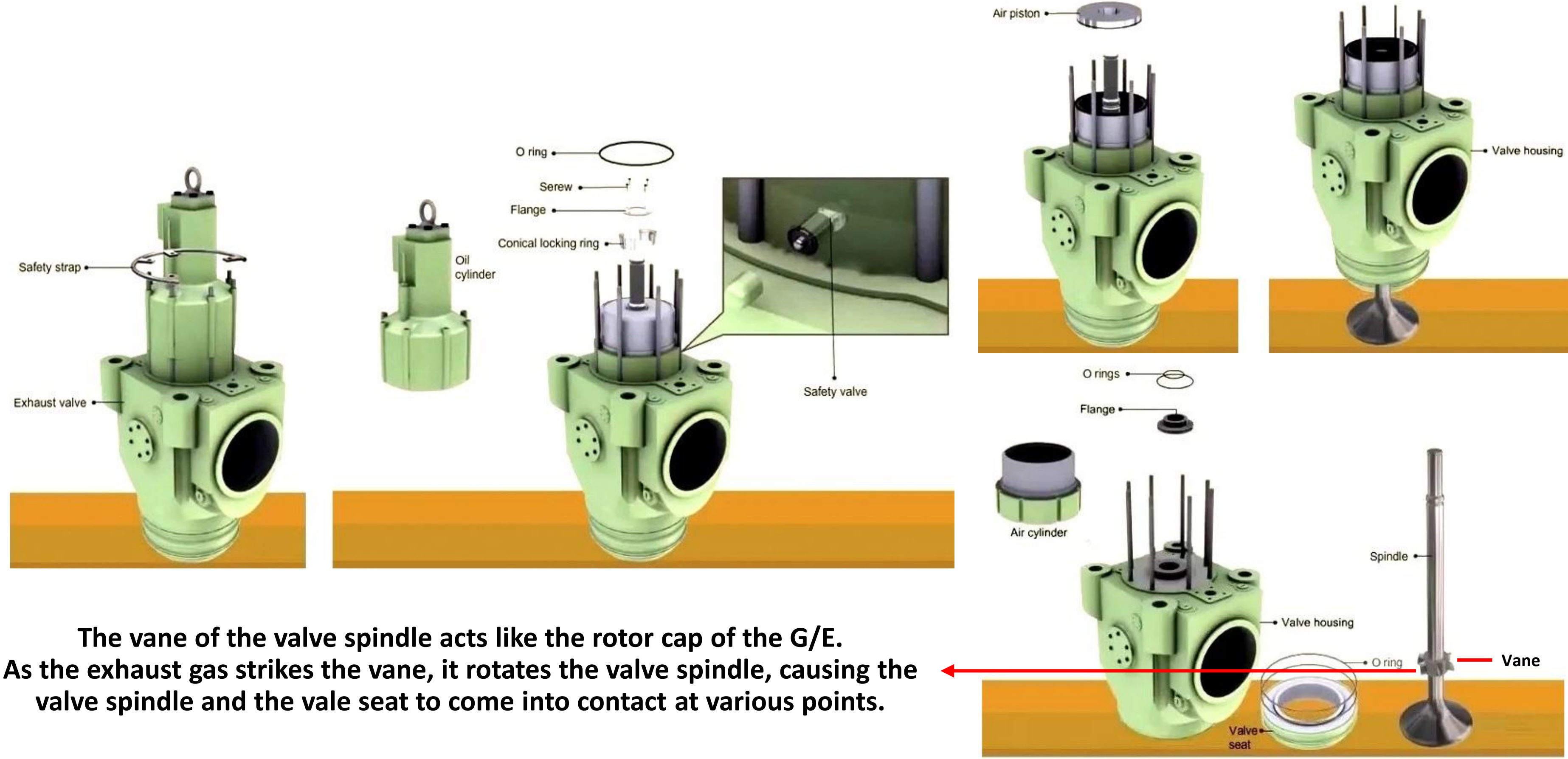
Valve spindles



Exhaust valve for M/E

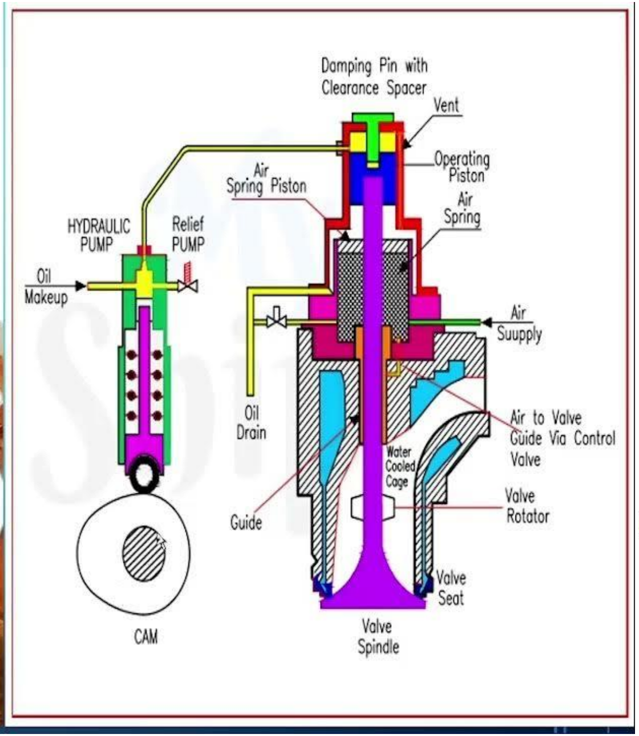


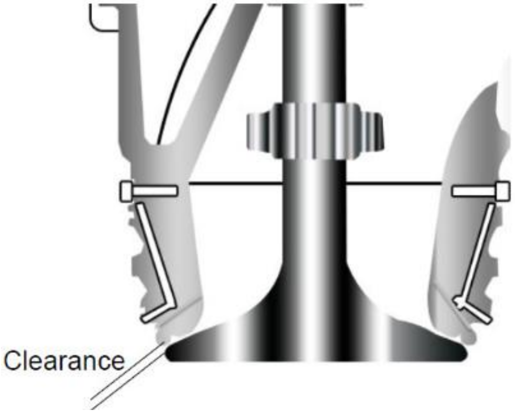
M/E exhaust valve



The vane of the valve spindle acts like the rotor cap of the G/E.
As the exhaust gas strikes the vane, it rotates the valve spindle, causing the valve spindle and the vale seat to come into contact at various points.

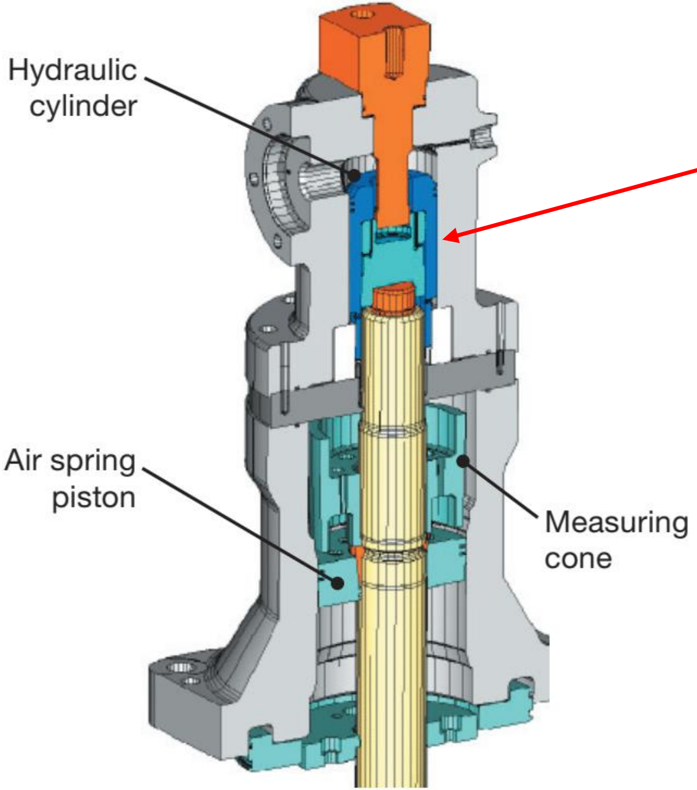
M/E exhaust valve

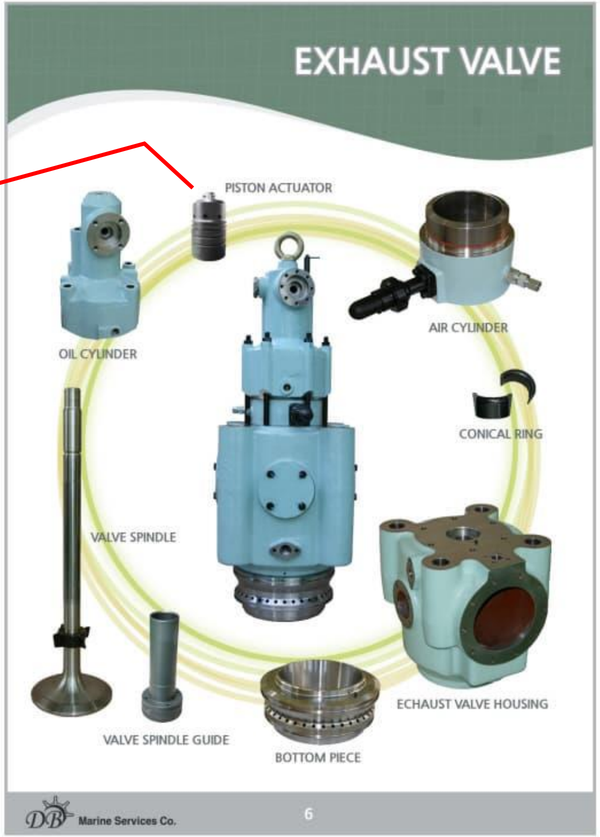




Exhaust valve for M/E

M/E exhaust valve





Exhaust valve for M/E

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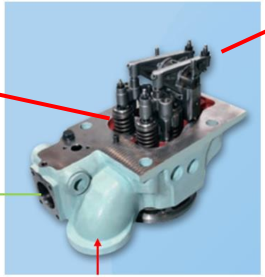
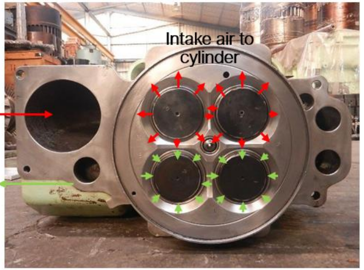
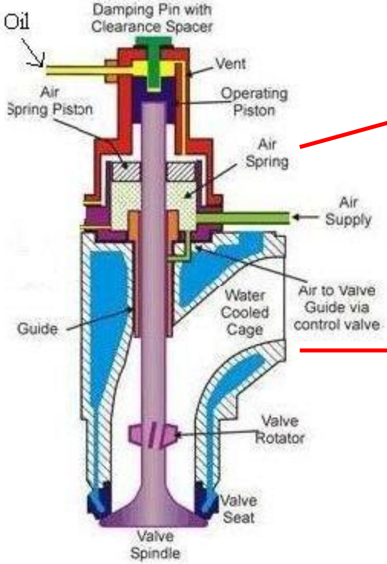
Difference between G/E and M/E

G/E	M/E
The two intake valves and two exhaust valves are closed by spring force and opened by the rocker arm operated by the cam of the camshaft.	The valve spindle of the exhaust valve is opened by pressurized hydraulic oil and closed by compressed air (it is called as spring air in the field).





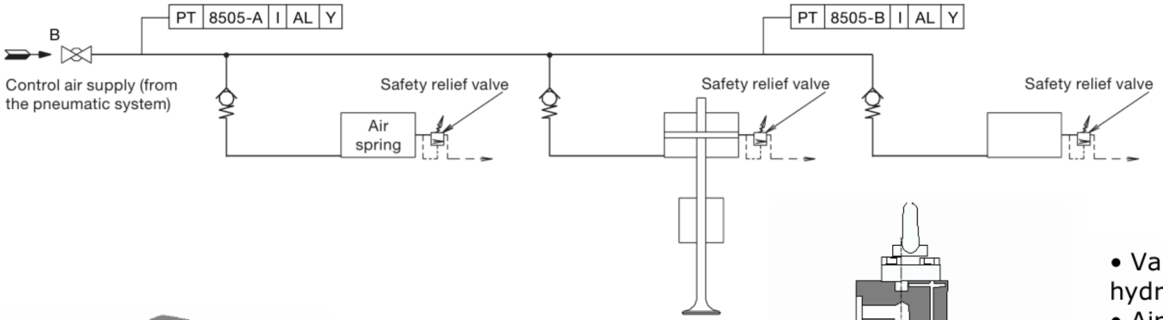
Valve spindles

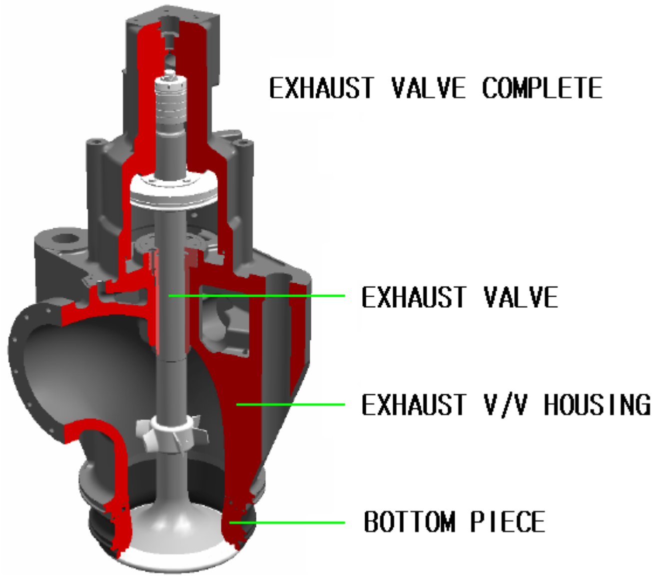


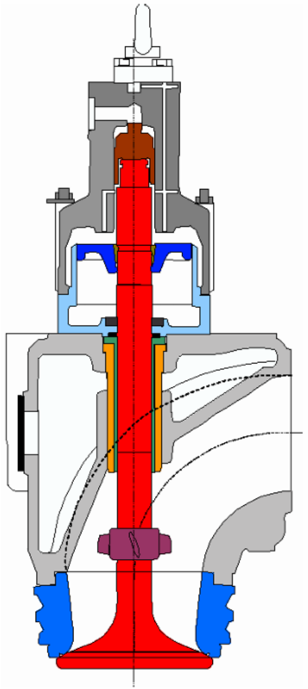
Air is worked as spring (it works to close exhaust valve – contact between valve spindle and valve seat)

Exhaust valve for M/E

M/E exhaust valve



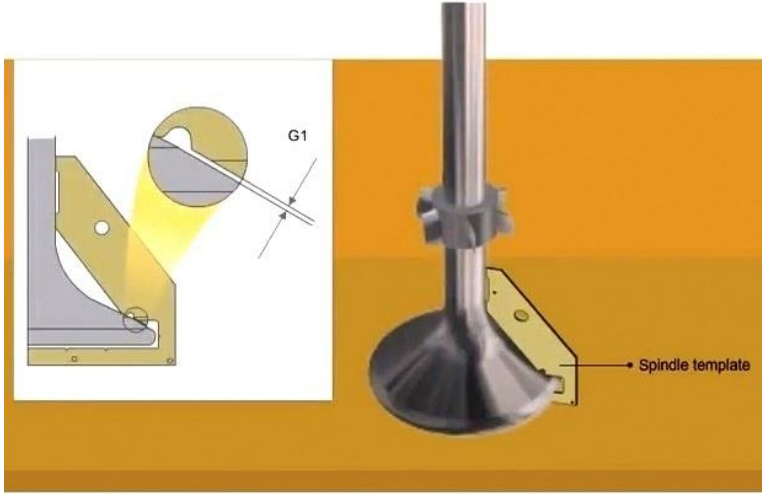


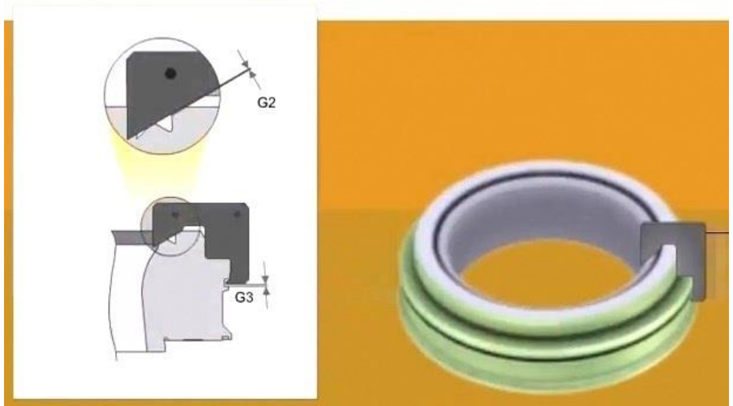


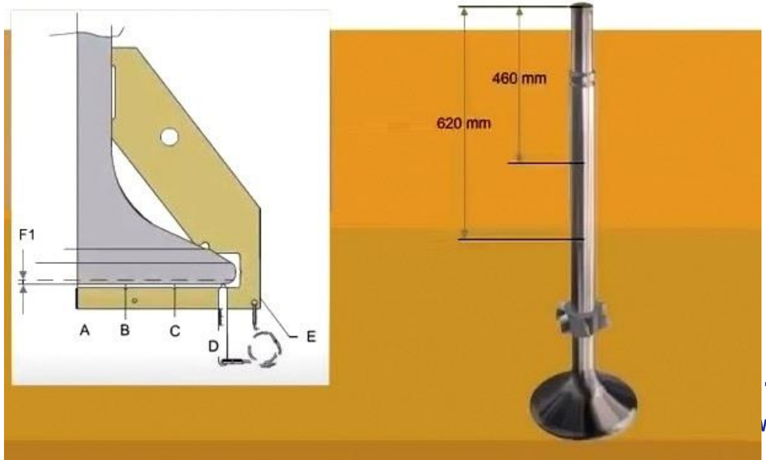
- Valve drive or hydraulic actuator
- Air cylinder
- Valve housing
- Valve guide
- Valve spindle
- Bottom piece or seat ring

Figure 01:
Main components of an exhaust valve

M/E exhaust valve







M/E exhaust valve

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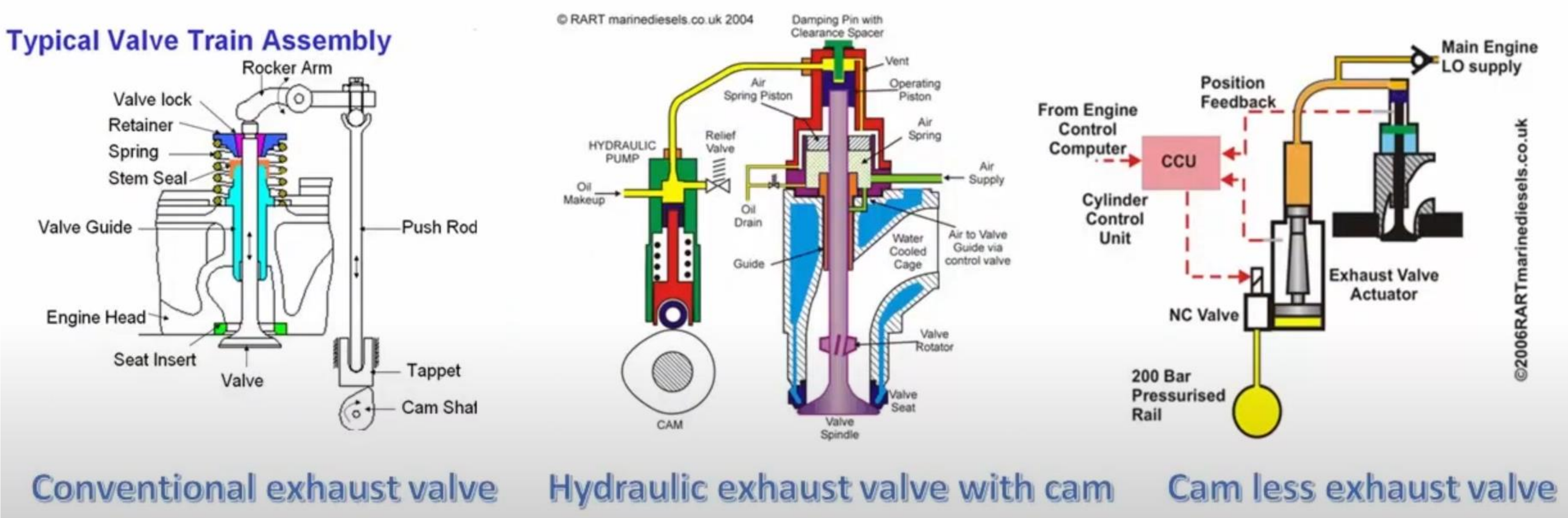
M/E exhaust valve

Difference between G/E and M/E

Old style **Current style**



G/E

M/E with camshaft
(mechanically driven)
MC type

M/E without camshaft
(electrically driven)
ME type

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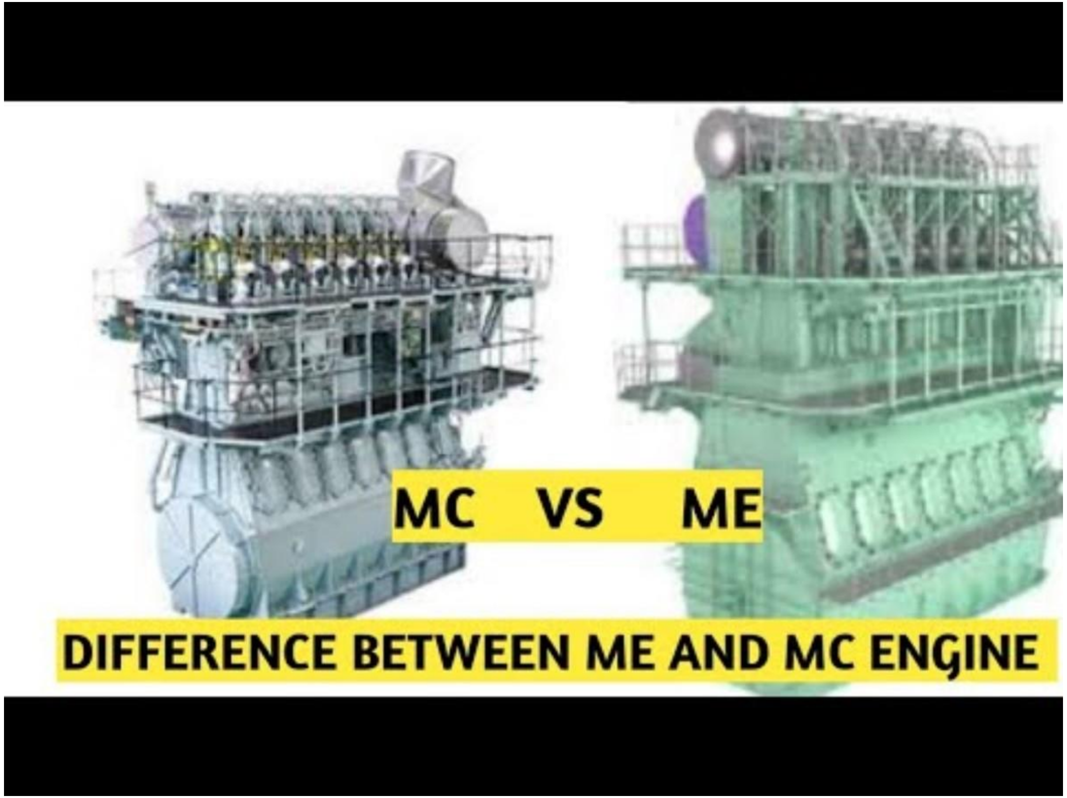
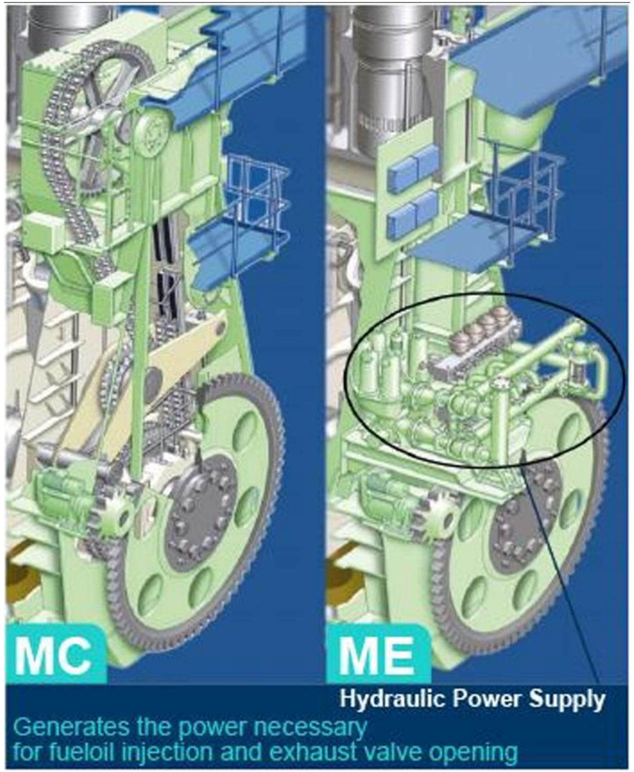
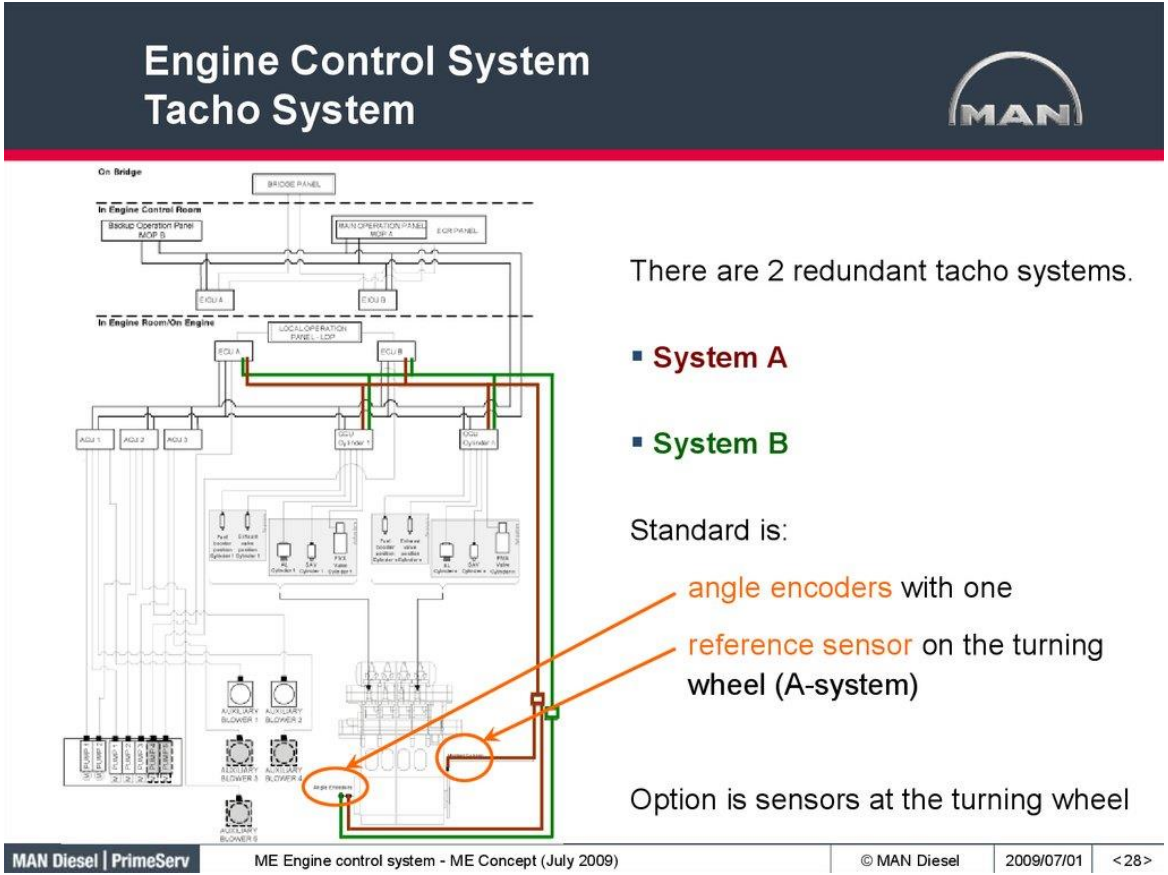
M/E (MAN B&W – ME type (not MC type))





M/E (ME type) – Tacho system

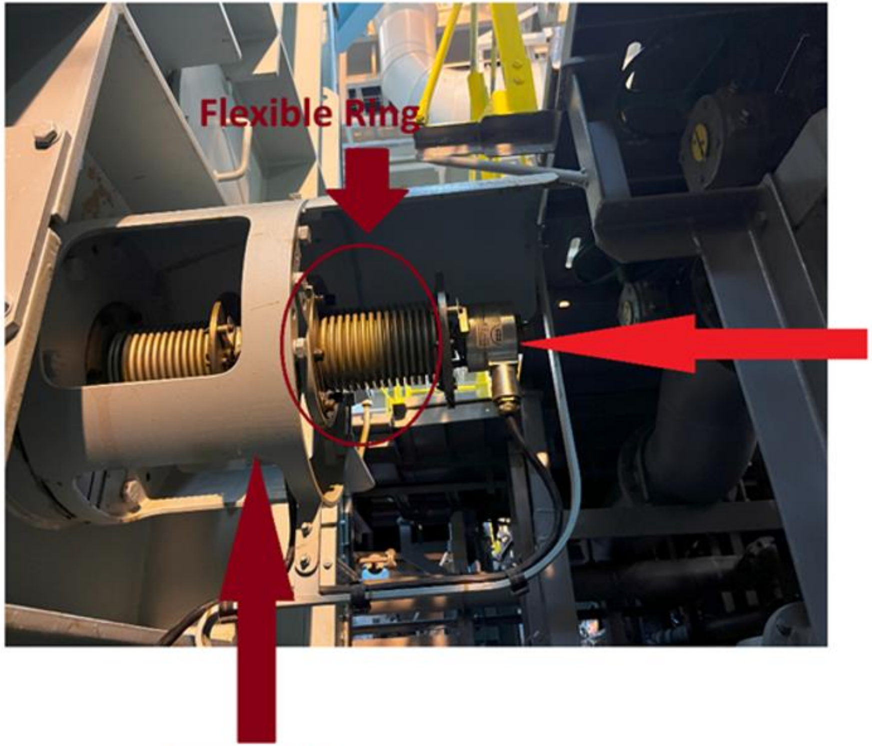


MC engines inject fuel and open exhaust valves via camshaft mechanisms.

ME engines use 1) angle encoders with 2) one reference sensor on the turning wheel to determine crankshaft position. Accordingly, the timing of fuel injection and exhaust valve opening is determined.

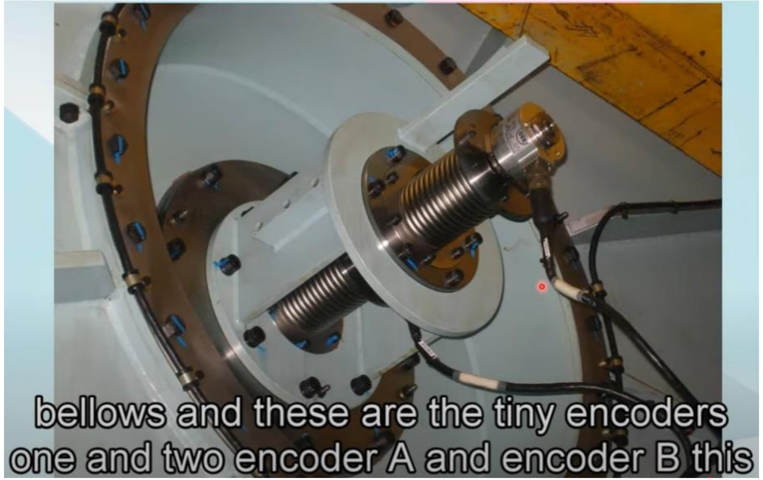
M/E (ME type) – Tacho system

M/E (ME type) – Tacho system



Encoder B

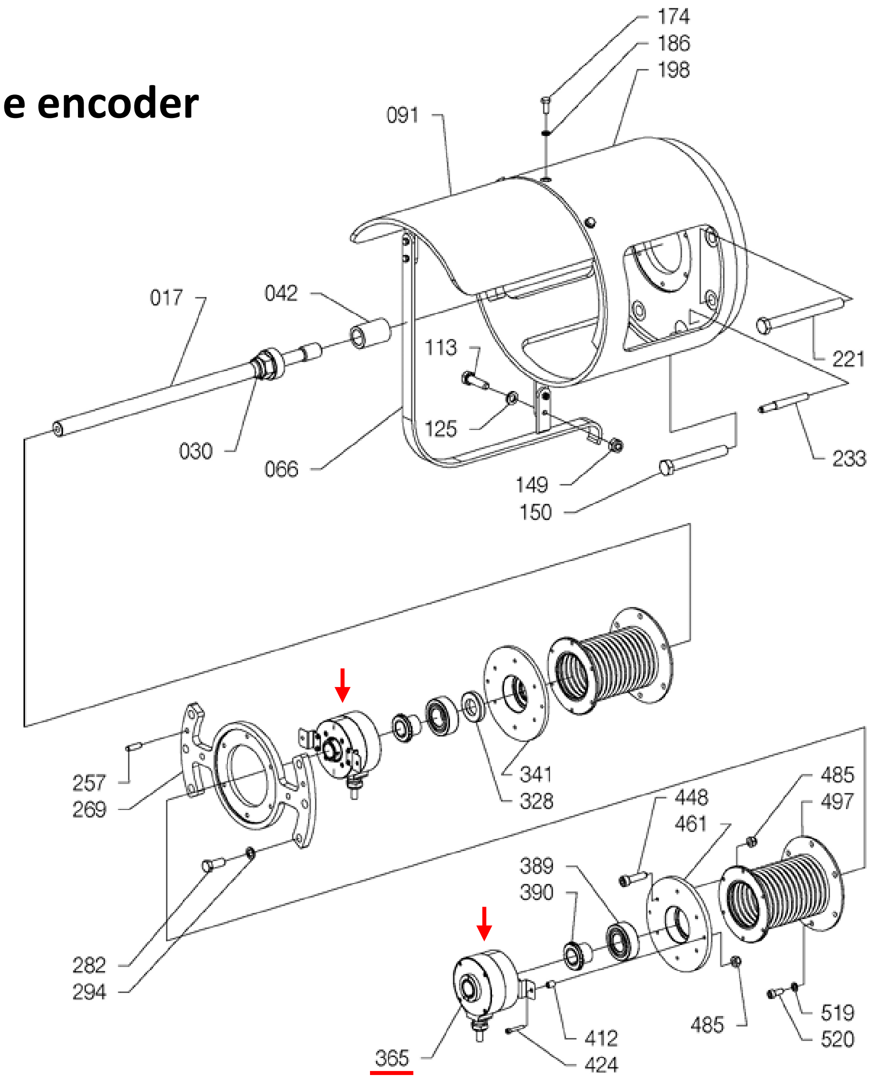
Encoder A



M/E (ME type) – Tacho system

https://www.youtube.com/live/Ed2xKhMCgMc?app=desktop

Angle encoder



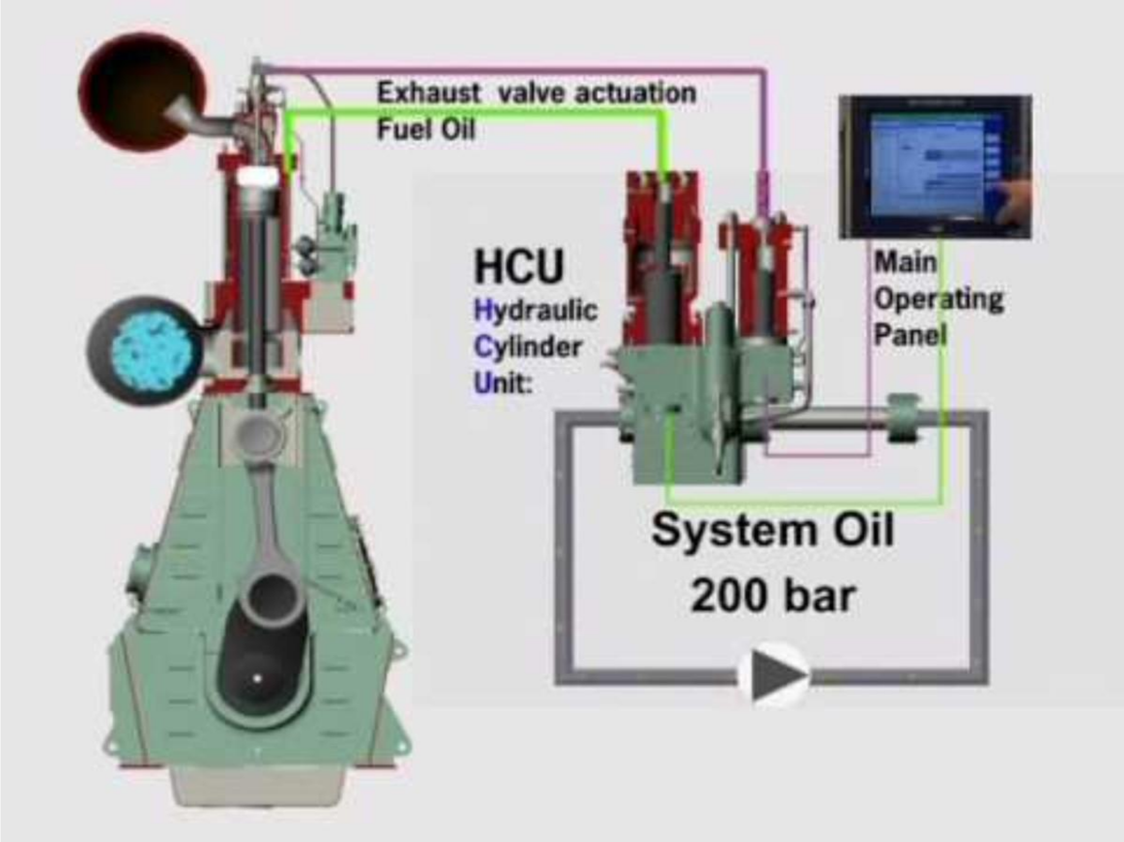
Item No.	Qty	Item Designation
017	-	Shaft
030	-	Sleeve
042	-	Distance pipe
066	-	Bracket
091	-	Bracket
113	-	Screw
125	-	Cam lock washer [1)]
149	-	Nut, self-locking
150	-	Screw
174	-	Screw
186	-	Cam lock washer [1)]
198	-	Bracket
221	-	Screw
233	-	Tapered dowel
257	-	Guide pin
269	-	Flange
282	-	Screw
294	-	Cam lock washer [1)]
328	-	Shaft seal
341	-	Flange bearing house
365	-	Angle encoder
389	-	Ball bearing, radial
390	-	Bearing sleeve ball brg.
412	-	Distance ring
424	-	Screw
448	-	Screw
461	-	Flange bearing house
485	-	Self-locking nut
497	-	Compensator
519	-	Spring lock
520	-	Screw

M/E

M/E flywheel

M/E (ME type) – Tacho system

M/E (MAN B&W – ME type (not MC type))



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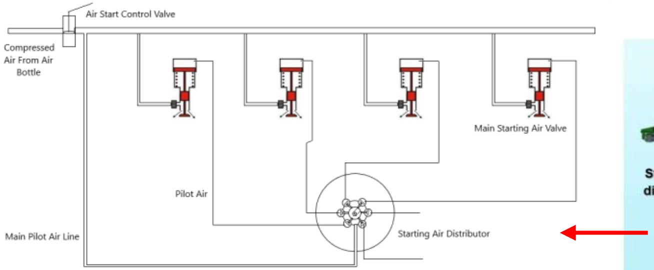
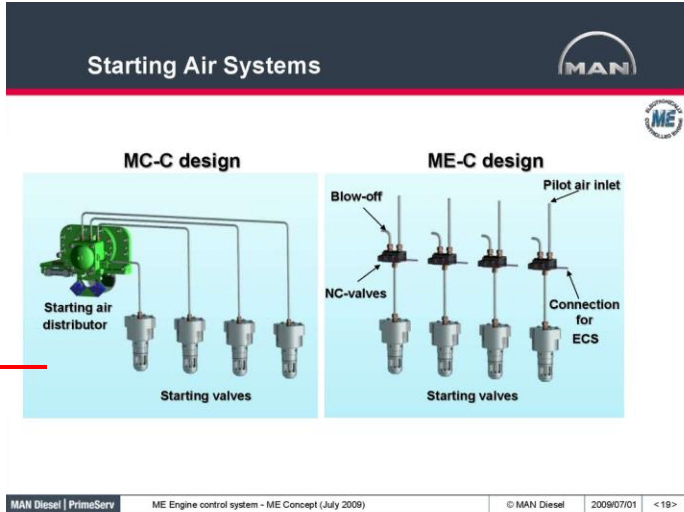
Difference between G/E and M/E

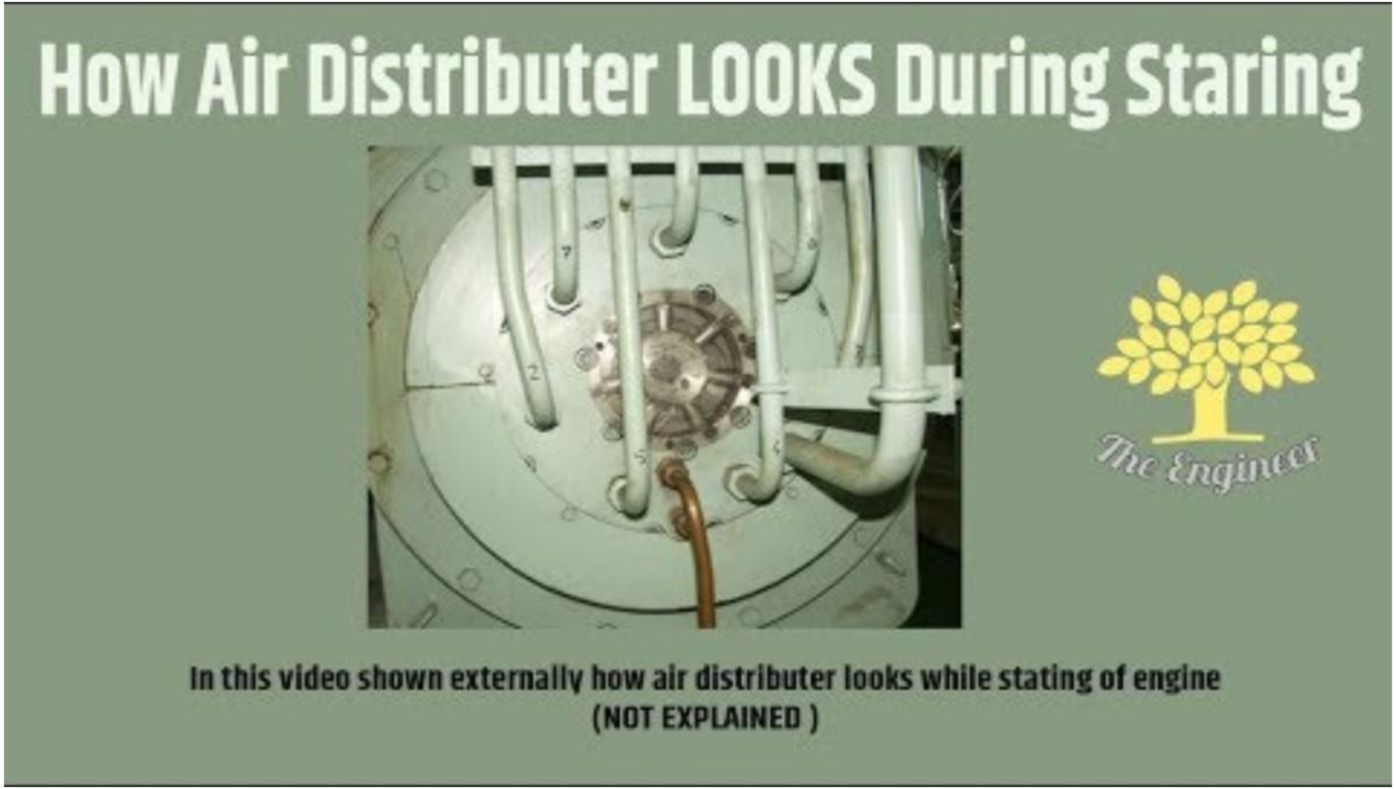
G/E	M/E
As the drive pinion of the air starter rotates by compressed air, the gear rim of the flywheel rotates, and as a result, the crankshaft connected to the flywheel rotates, starting the G/E.	The pressurized air in the air reservoir tank is injected into the cylinder by the air starting valve according to the ignition sequence, causing the piston to reciprocate and rotate the crankshaft, thereby starting the M/E.

Example of an air starter installed in a G/E

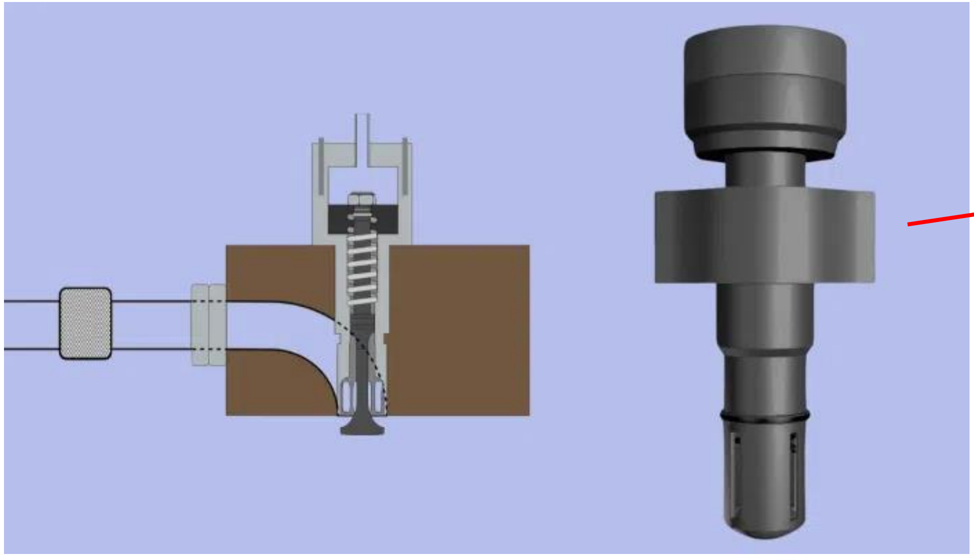


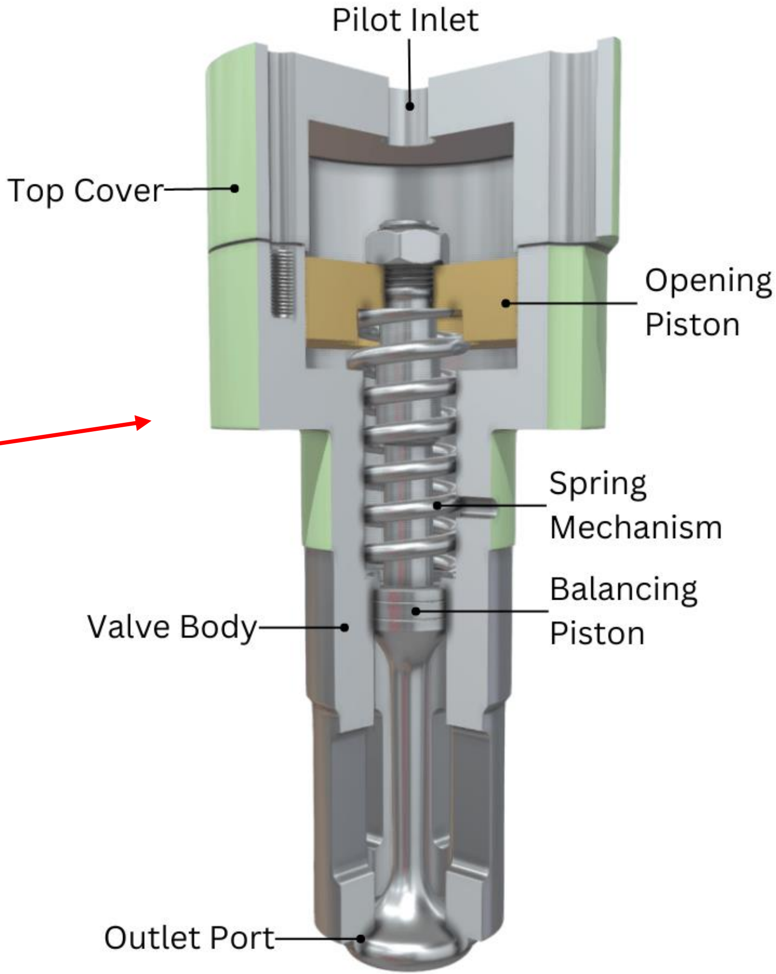


M/E air distributor



M/E air starting valve

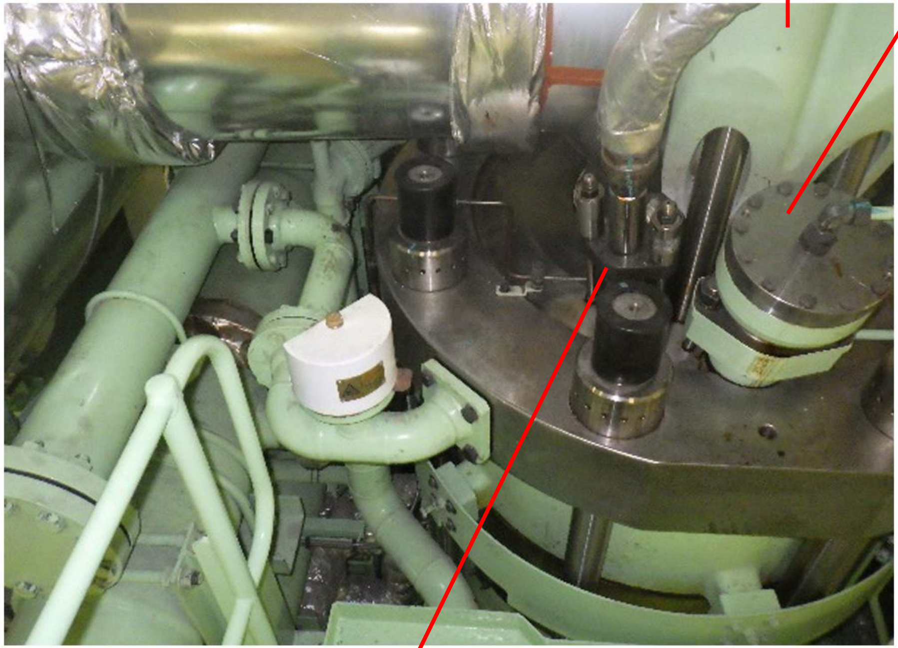




M/E air starting valve

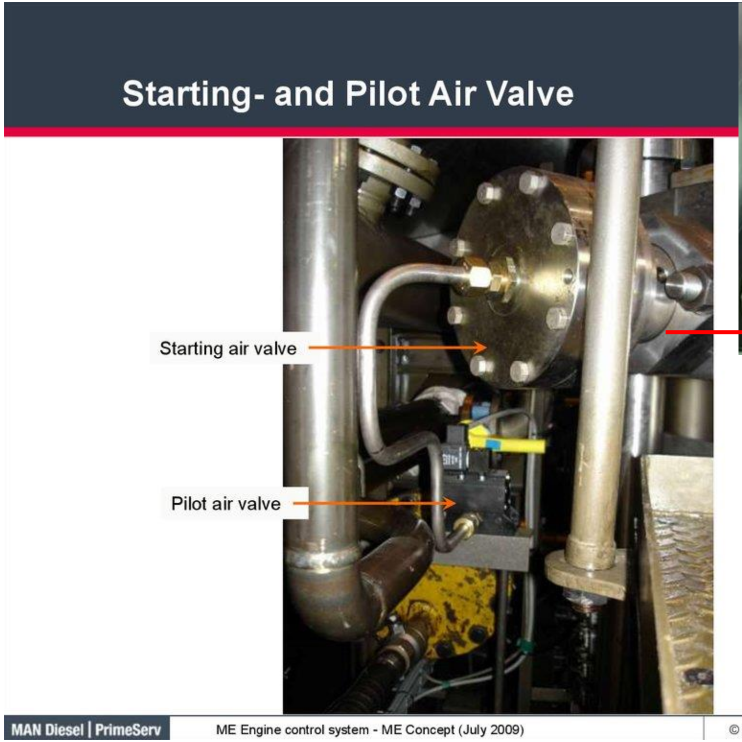
Air
starting
valve

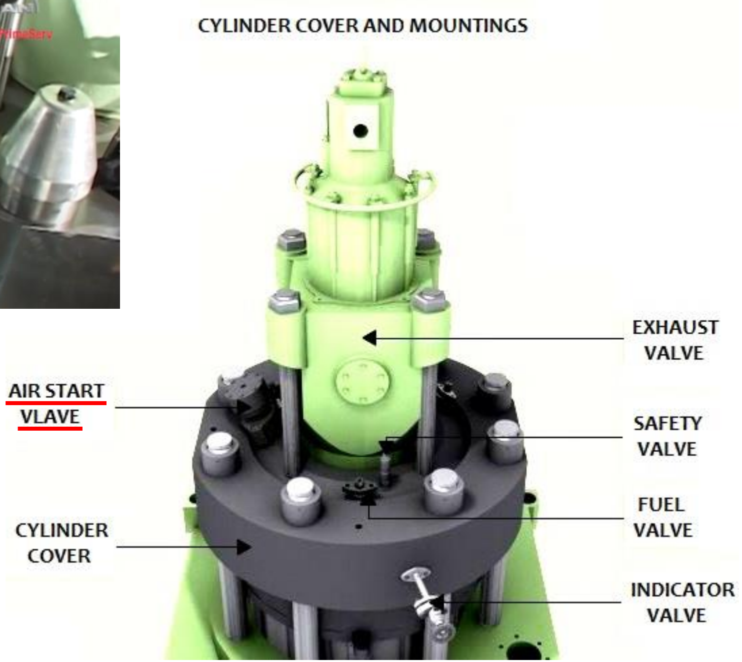
Exhaust valve



Fuel
injection
valve

M/E air starting valve





The air starting valve of some M/Es is installed on the side of the cylinder cover. However, it is generally installed on top of the cylinder cover. It is not the location of the air starting valve that is important, but its function.

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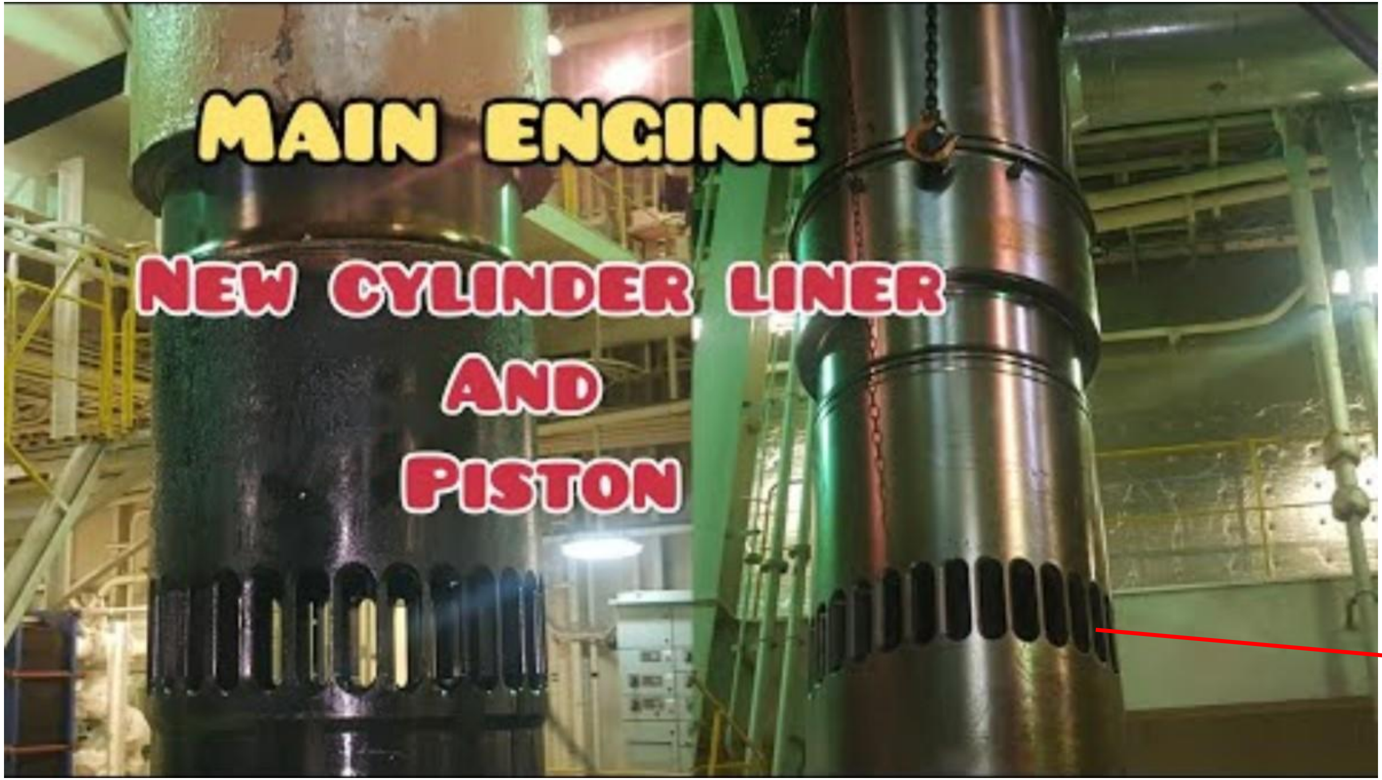
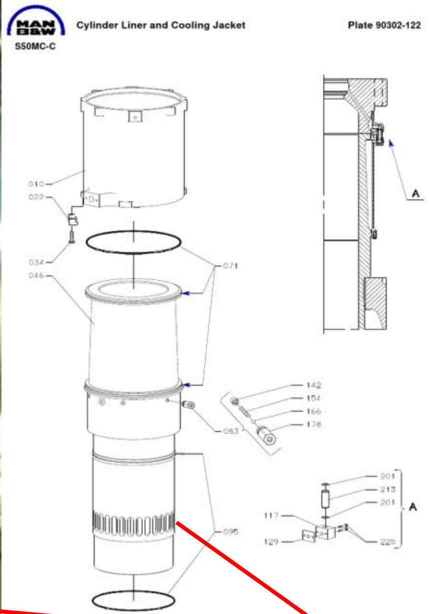
M/E cylinder cover and cylinder liner



2:10 – Cylinder cover
4:08 – Cylinder liner

Cylinder liner



Scavenge port

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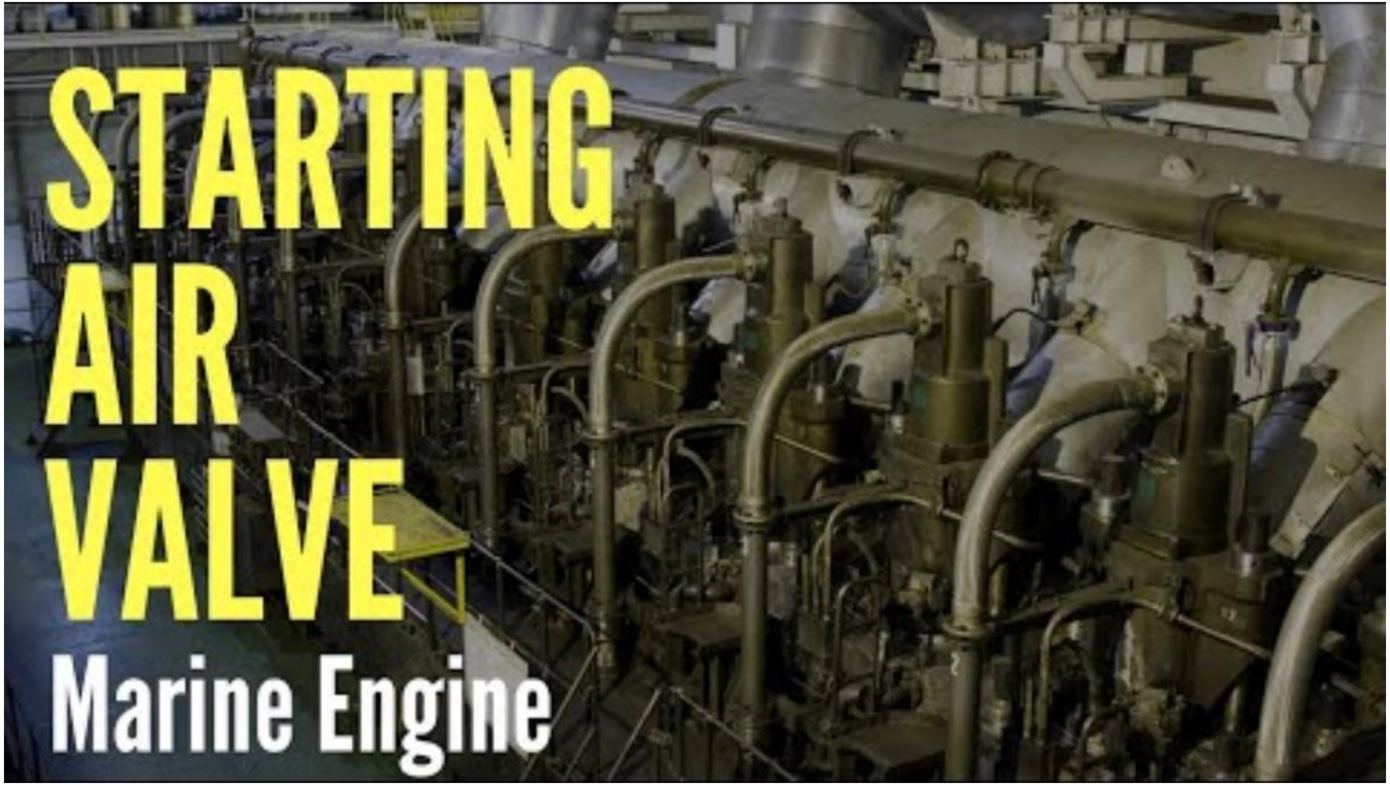
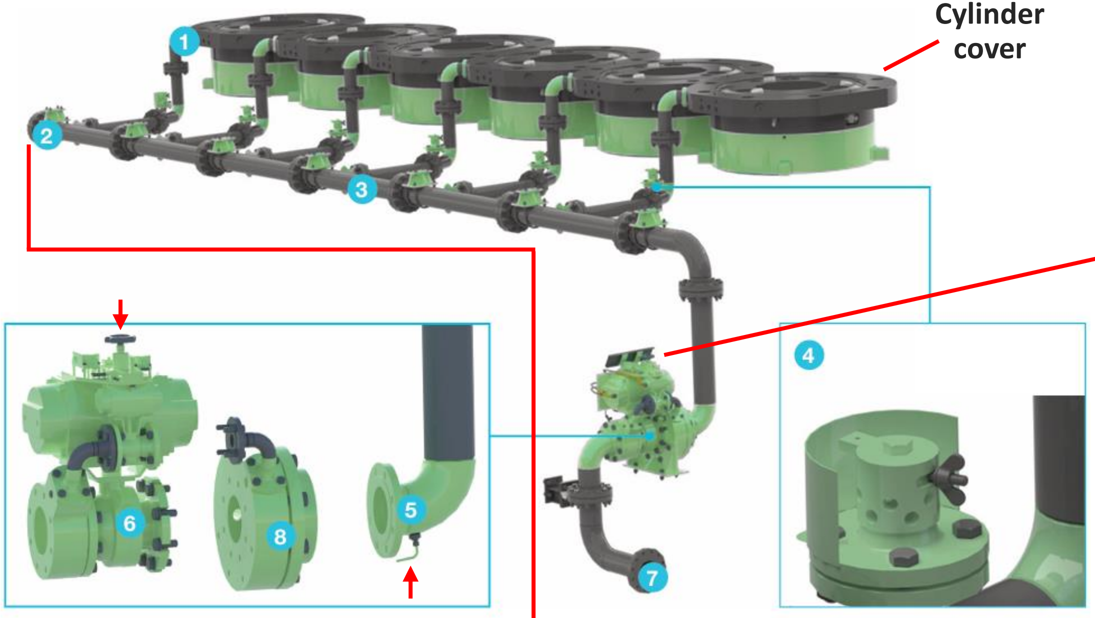
M/E air starting valve



0:00 – MAN B&W

3:16 – Wartsila

M/E air starting system



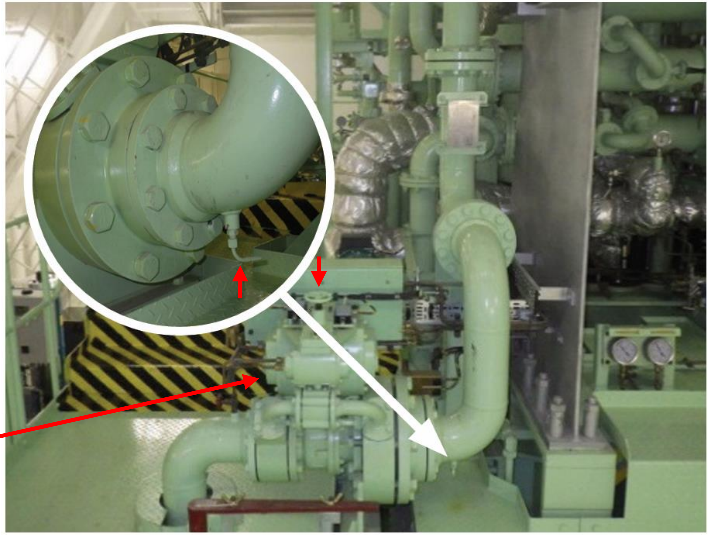


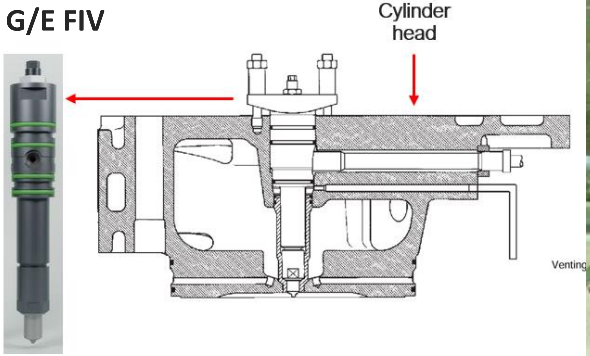
Fig. 2: Main starting air valve

1. Starting air valve
2. Drain/venting pipe
3. Starting air manifold
4. Bursting cap and bursting disc
5. Drain/venting pipe
6. Main starting valve
7. Starting air supply
8. Non-return valve

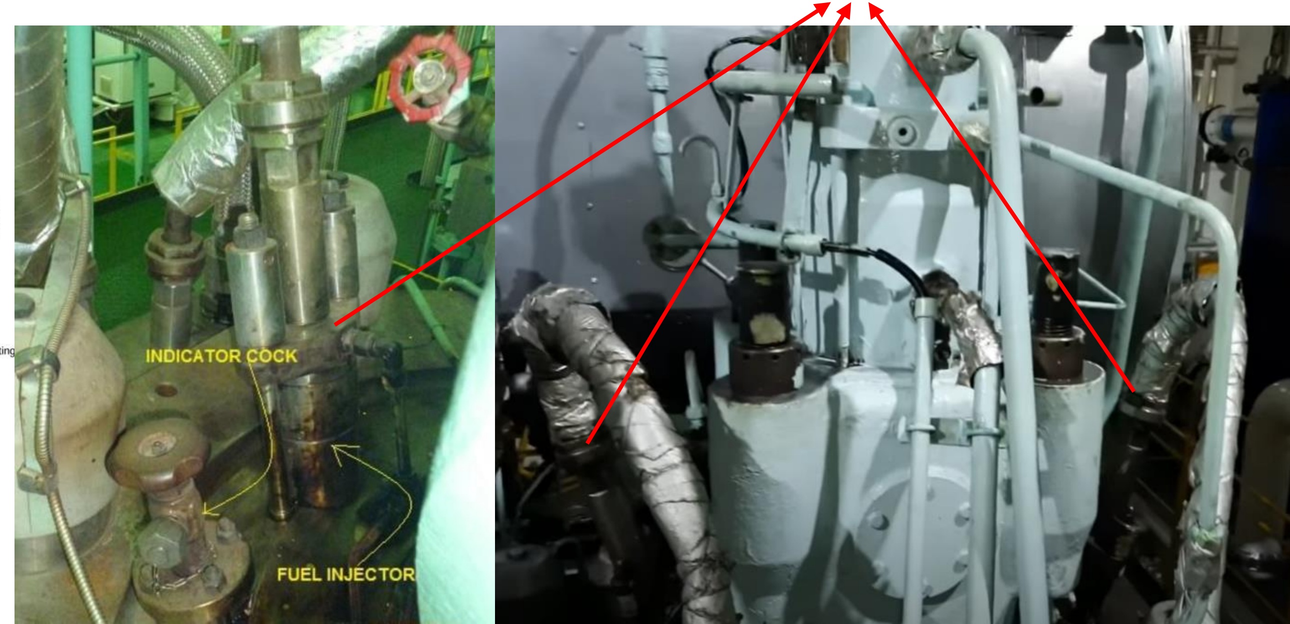
Difference between G/E and M/E

G/E	M/E
Typically, a G/E is equipped with one fuel injection valve (FIV).	Typically, M/Es are equipped with 2 to 3 fuel injection valves.

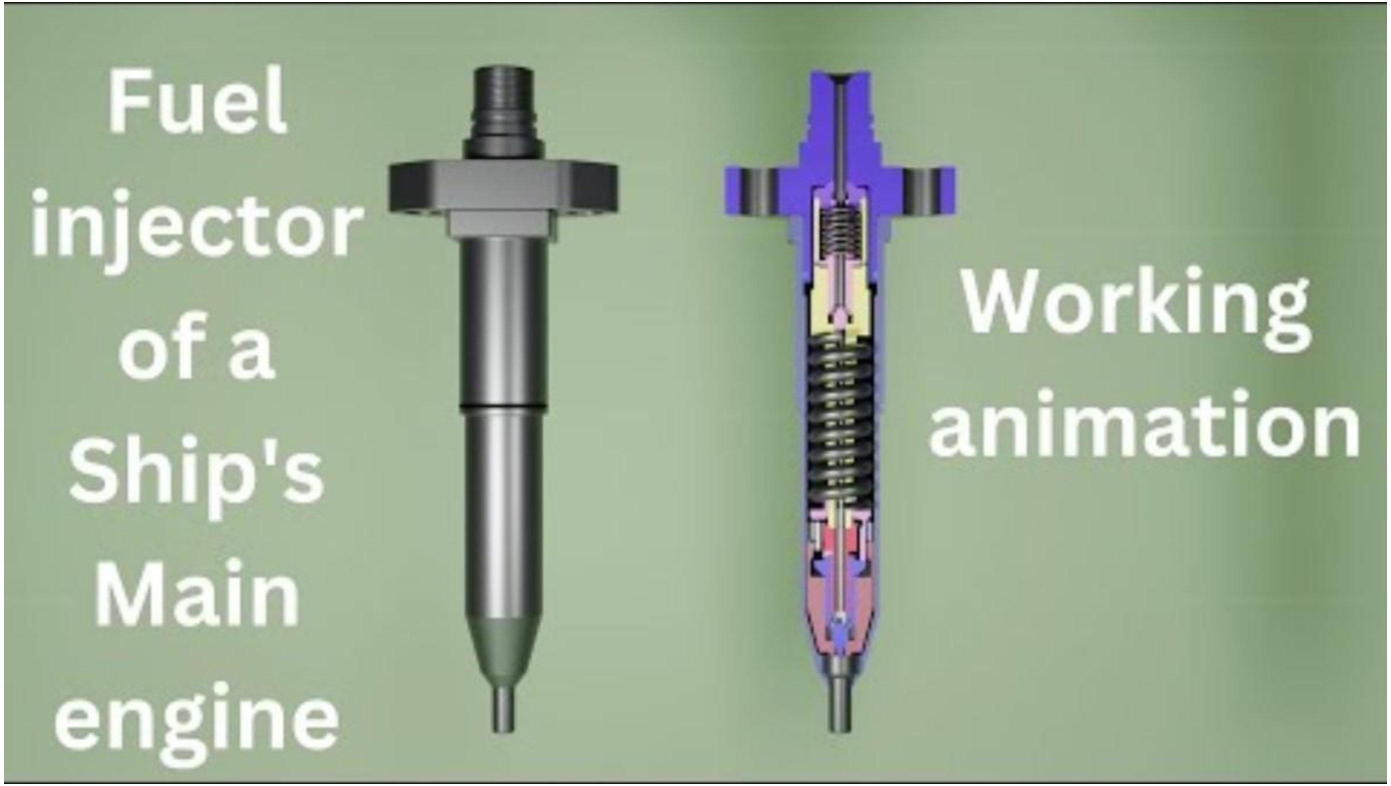
G/E FIV



M/E FIV



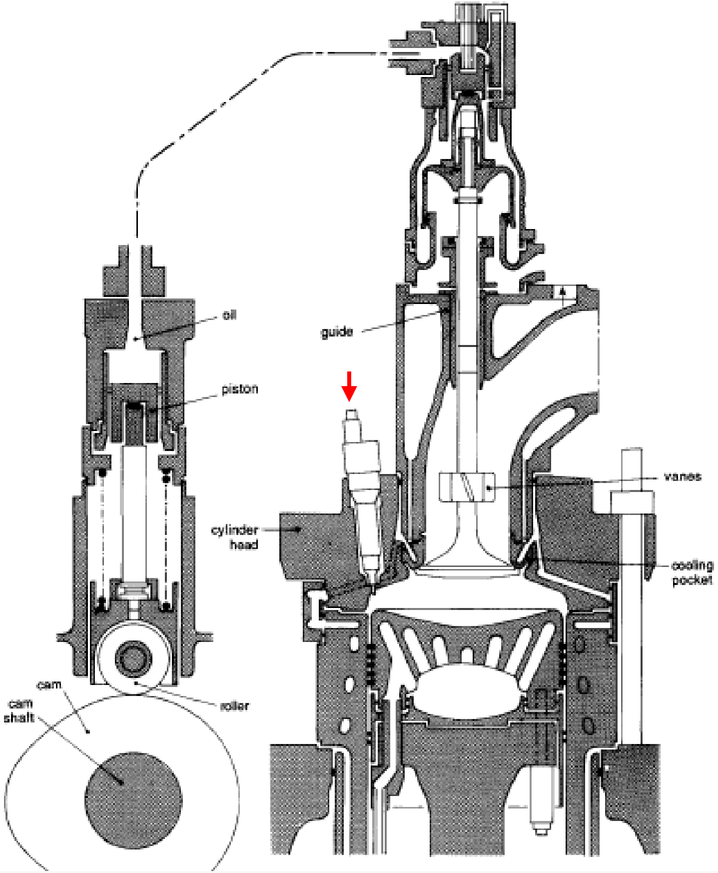
Fuel injection valve (FIV)



Fuel injection valve (FIV)

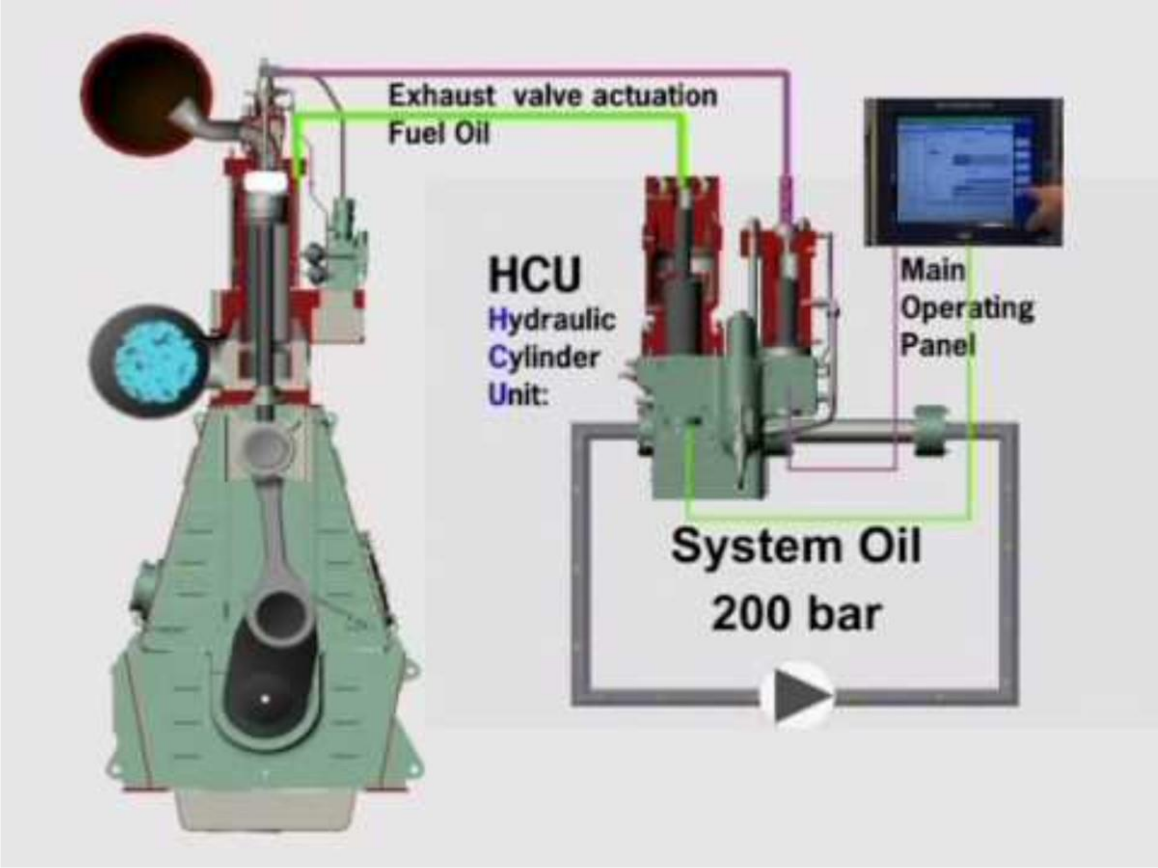
Fuel injection valve (FIV)

M/E with camshaft (mechanically driven) MC type



High pressure pipe and FIV

M/E (MAN B&W – ME type (not MC type))



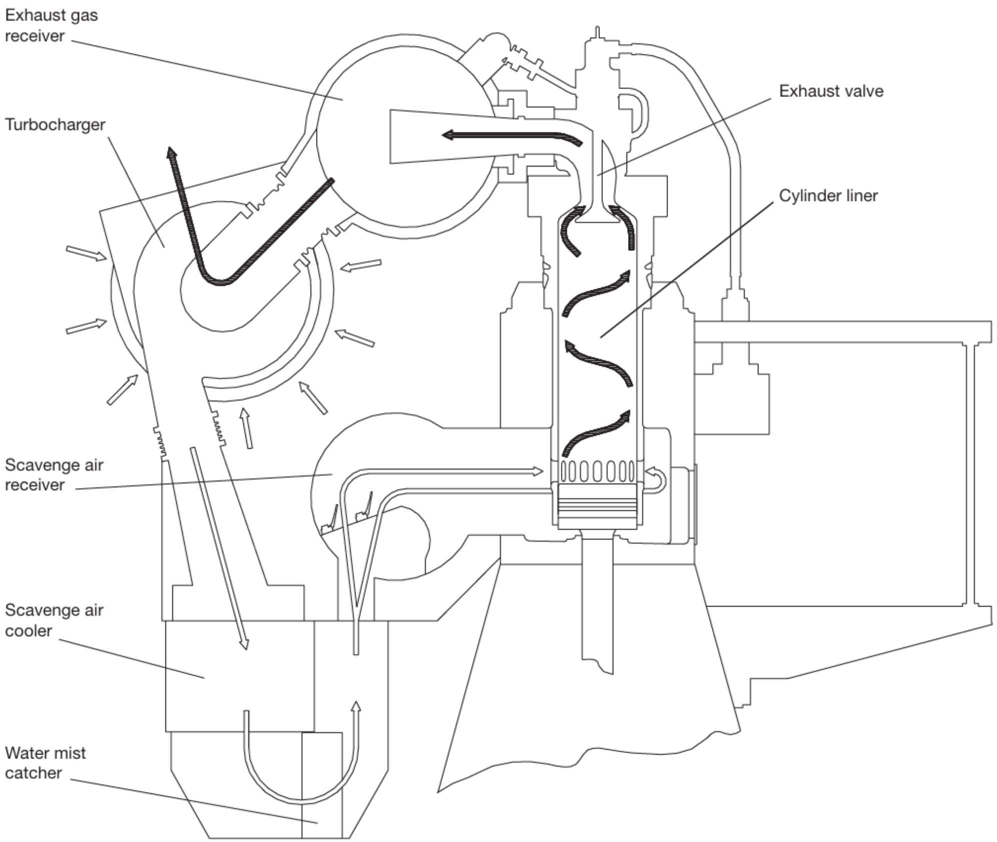
Scavenging system of M/E

Outside air sucked into the compressor side of the T/C is compressed and sent to the air cooler.

The cooled air from the air cooler passes through the scavenge air receiver and enters the cylinder liner through the scavenge port of the cylinder liner.

Compressed cooling air entering the cylinder liner pushes the exhaust gas toward the exhaust gas receiver as the piston moves upward. (At this time the exhaust valve is open.)

The exhaust gas in the exhaust gas receiver goes to the turbine side of the T/C, rotates the turbine wheel, and is discharged into the atmosphere through a funnel.

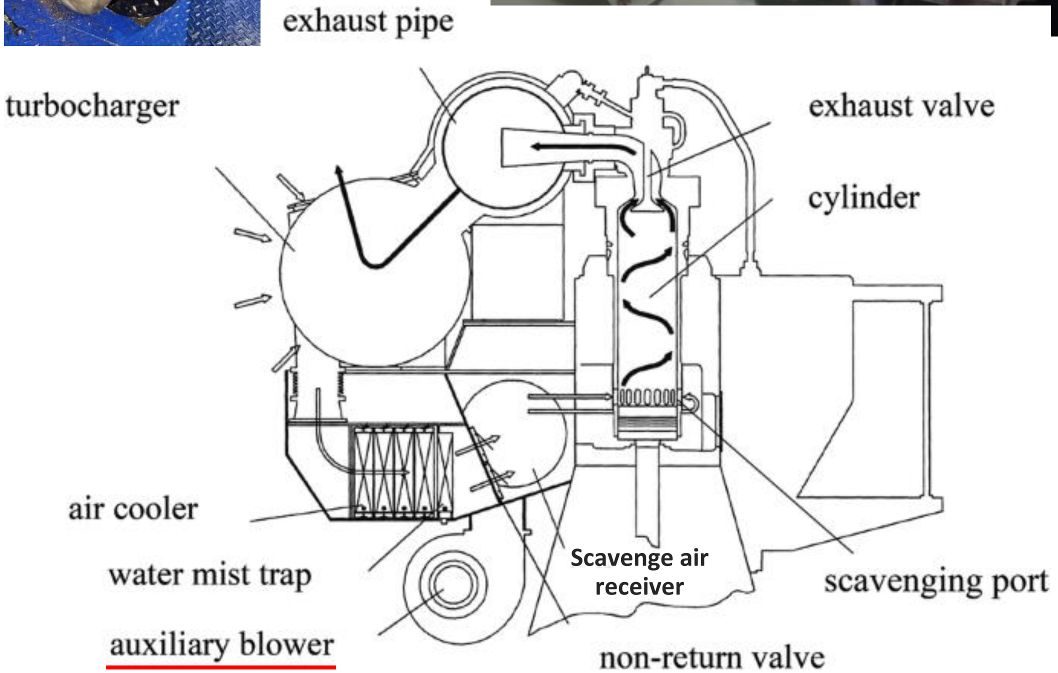


M/E auxiliary blower

At the beginning of M/E operation, the heat energy of the exhaust gas is not high, so the rotational power of the T/C turbine is insufficient, and accordingly, the pressure of the charge air by the compressor is insufficient.

This phenomenon creates black smoke because the compressed air is insufficient compared to the injected fuel.

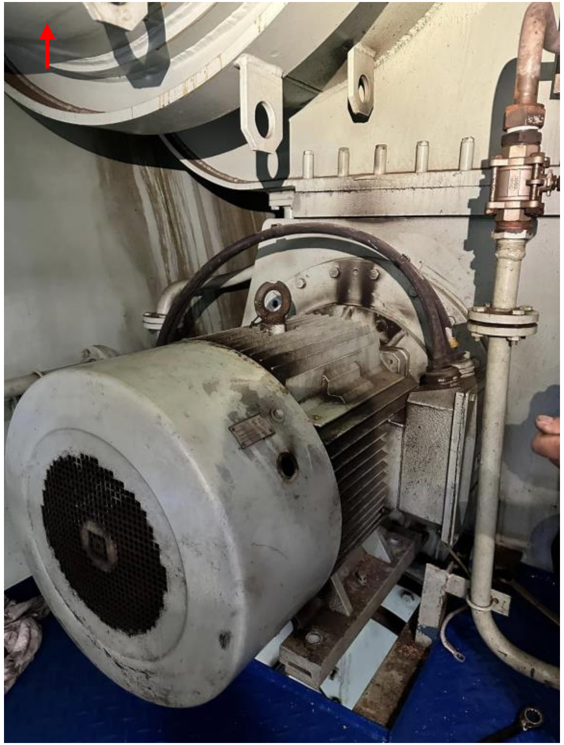
Therefore, at the beginning of M/E operation, M/E auxiliary blowers are activated to increase the pressure of the charge air.

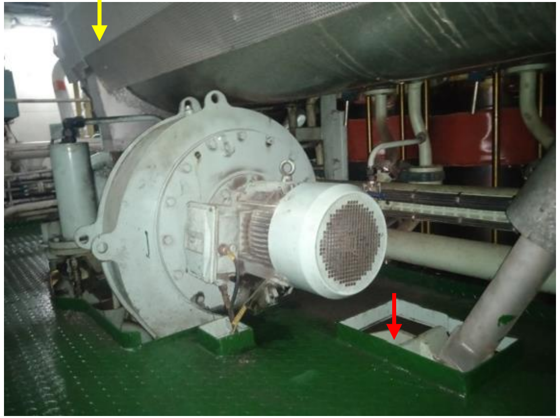


M/E auxiliary blower

Scavenge air receiver :

Exhaust manifold





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M/E T/C


ME T/C
ABB
MAIN ENGINE TURBO CHARGERS
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Scavenging system of M/E

Lantern space

Example of scavenging air receiver non-return valves

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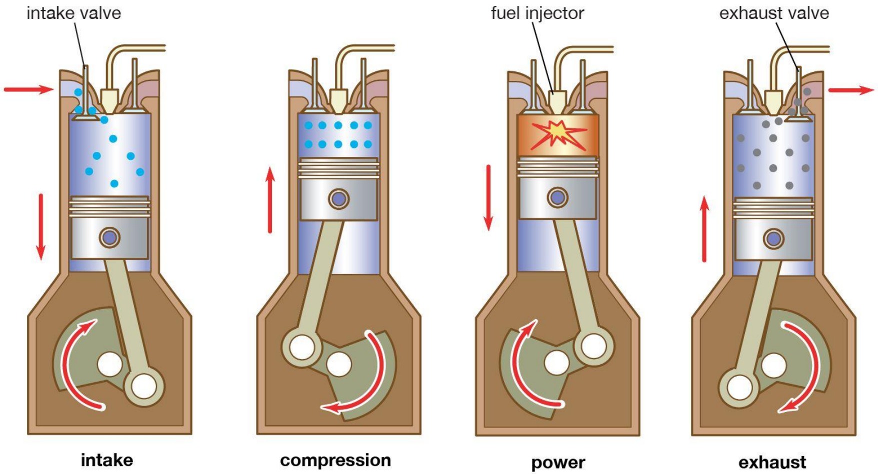
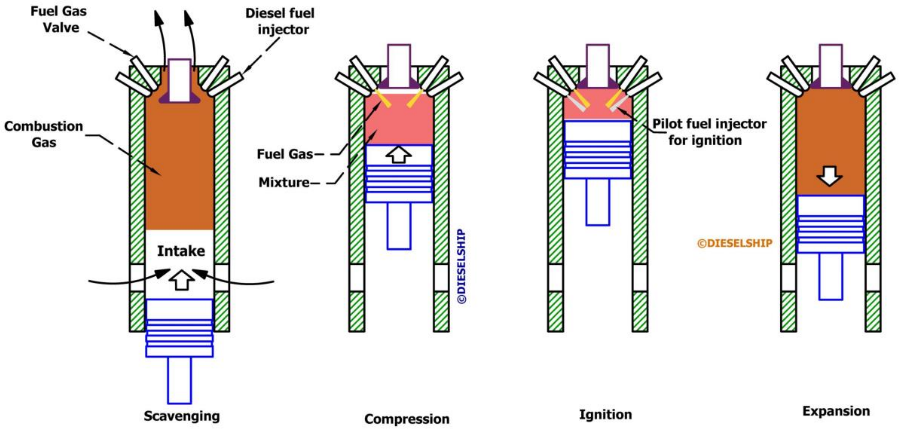
Difference between G/E and M/E

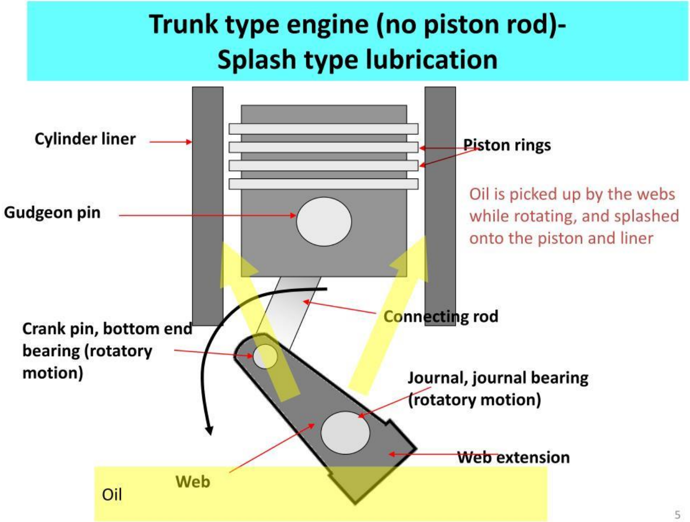
G/E	M/E
The intake of intake air and the discharge of exhaust gas occur in separate stroke processes.	The intake of intake air and the discharge of exhaust gas occur in same stroke process. As a result, mixing of intake air and exhaust gas is inevitable.

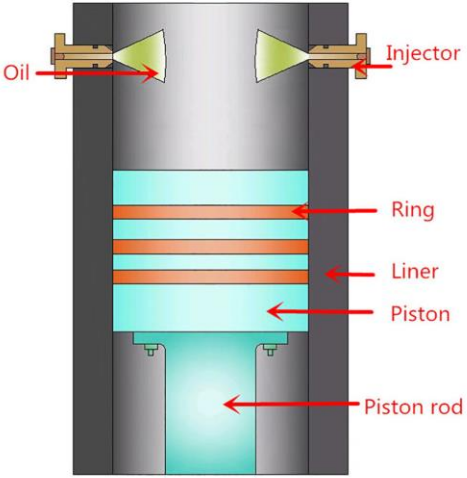


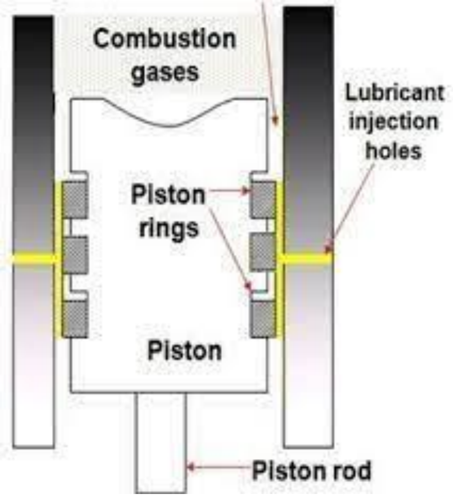


Difference between G/E and M/E

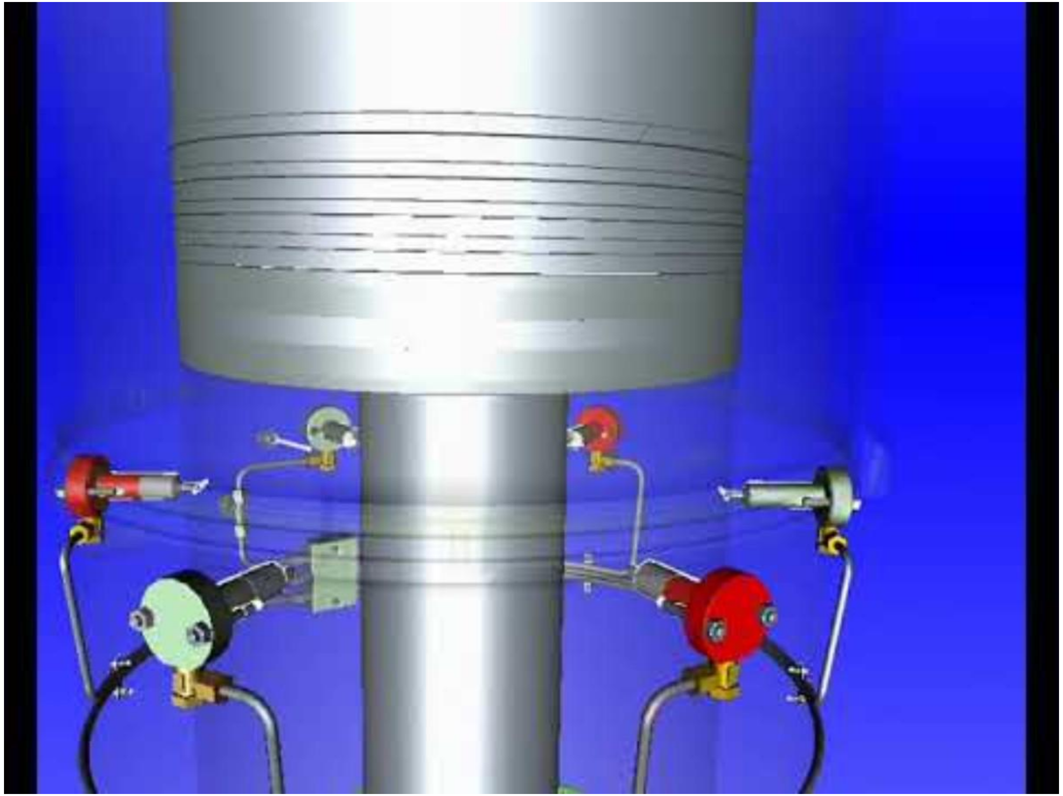
G/E	M/E
Lubrication between the cylinder liner and piston rings of the G/E is achieved by the splash effect caused by the rotation of the crankshaft and oil vapor.	Lubrication between the cylinder liner and piston ring of the M/E is achieved by lubricating oil sprayed directly onto the cylinder liner.

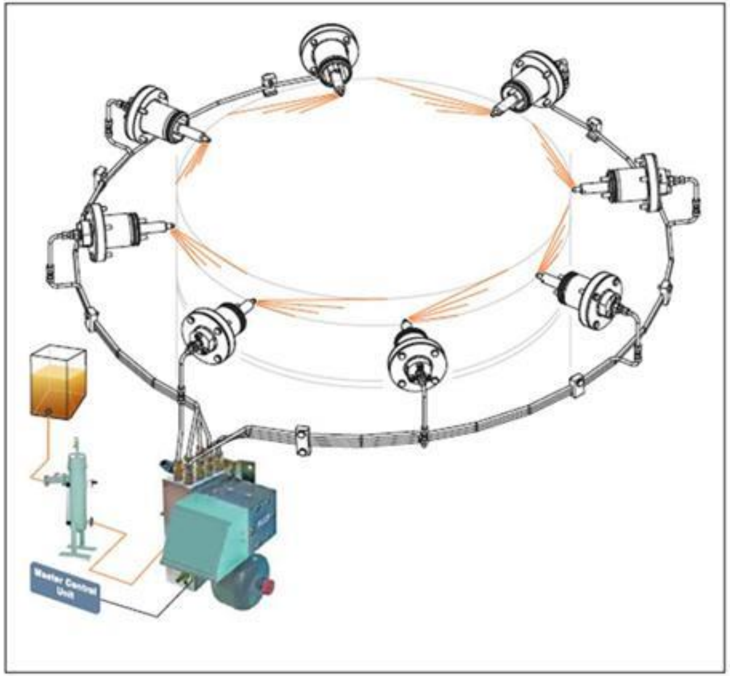






Difference between G/E and M/E





Alpha cylinder lubricator

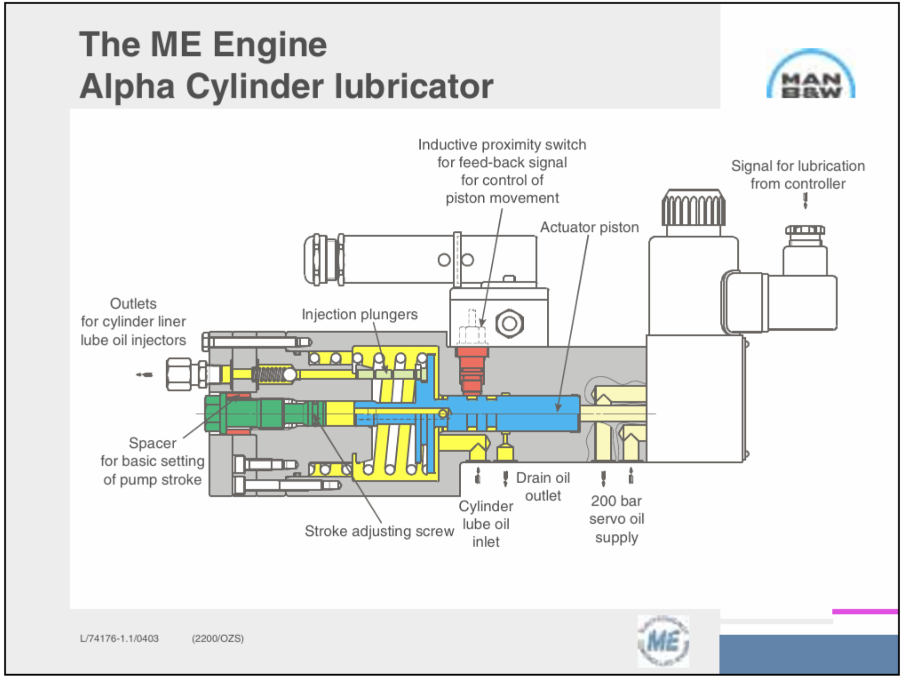


Fig. 17: Alpha Lubricator for ME Engine

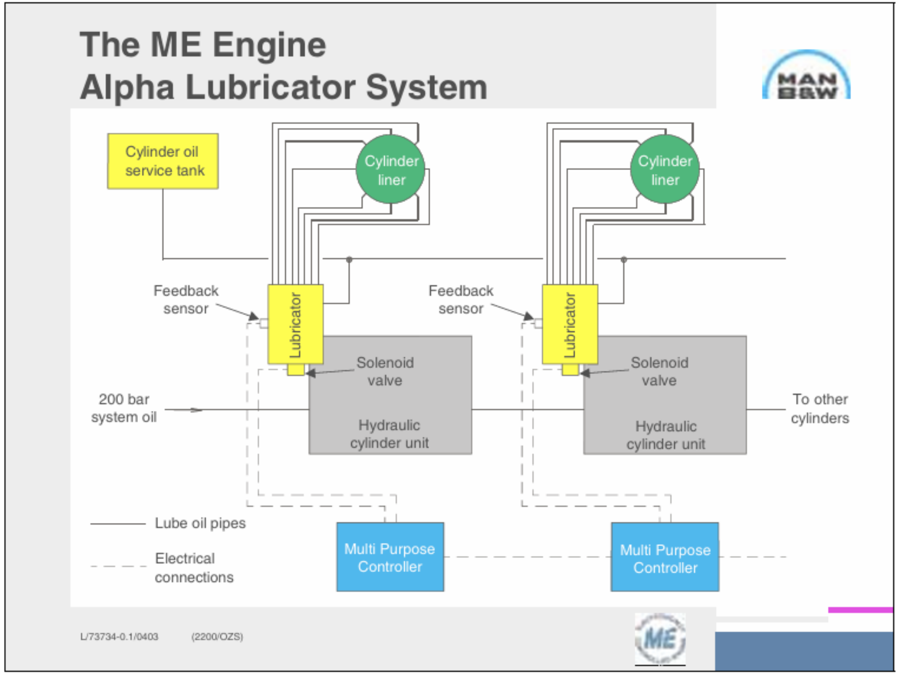
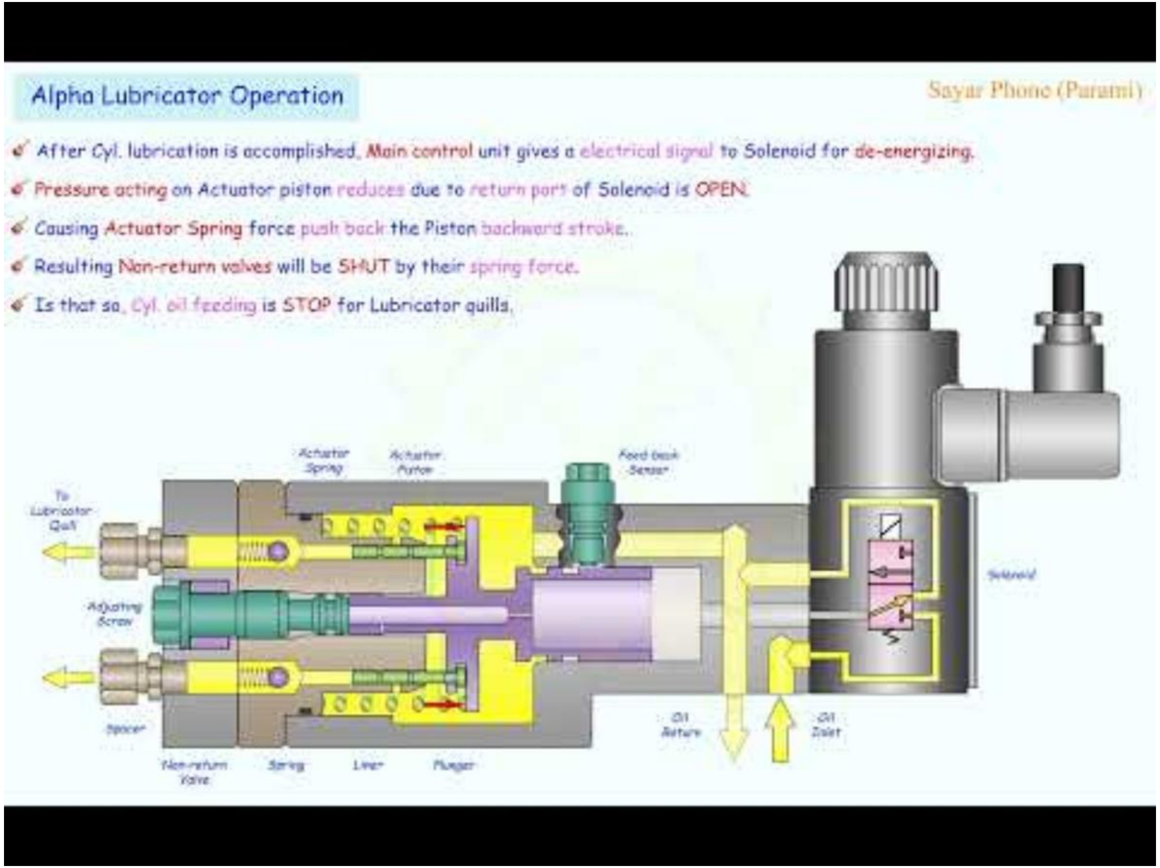


Fig. 18: Alpha Lubricator System for ME

Alpha cylinder lubricator

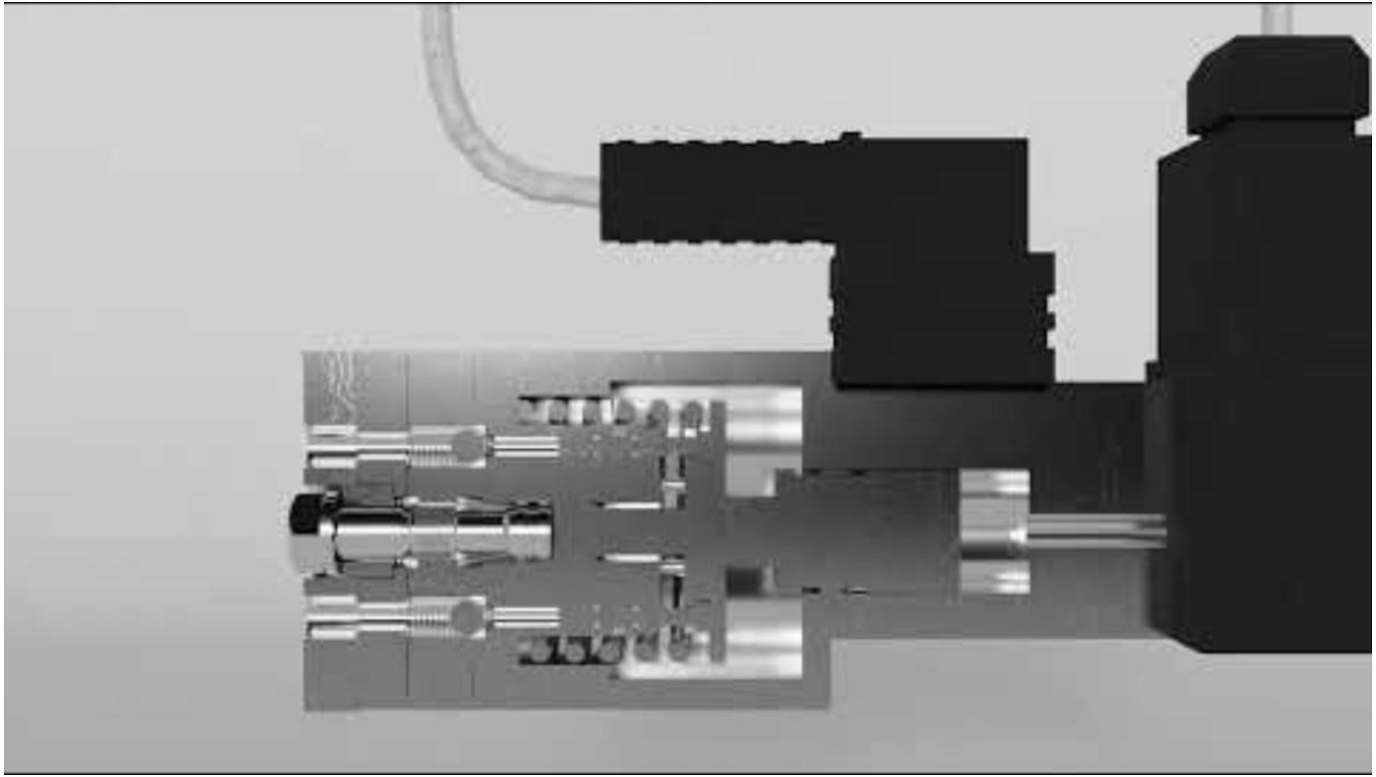


Alpha cylinder lubricator

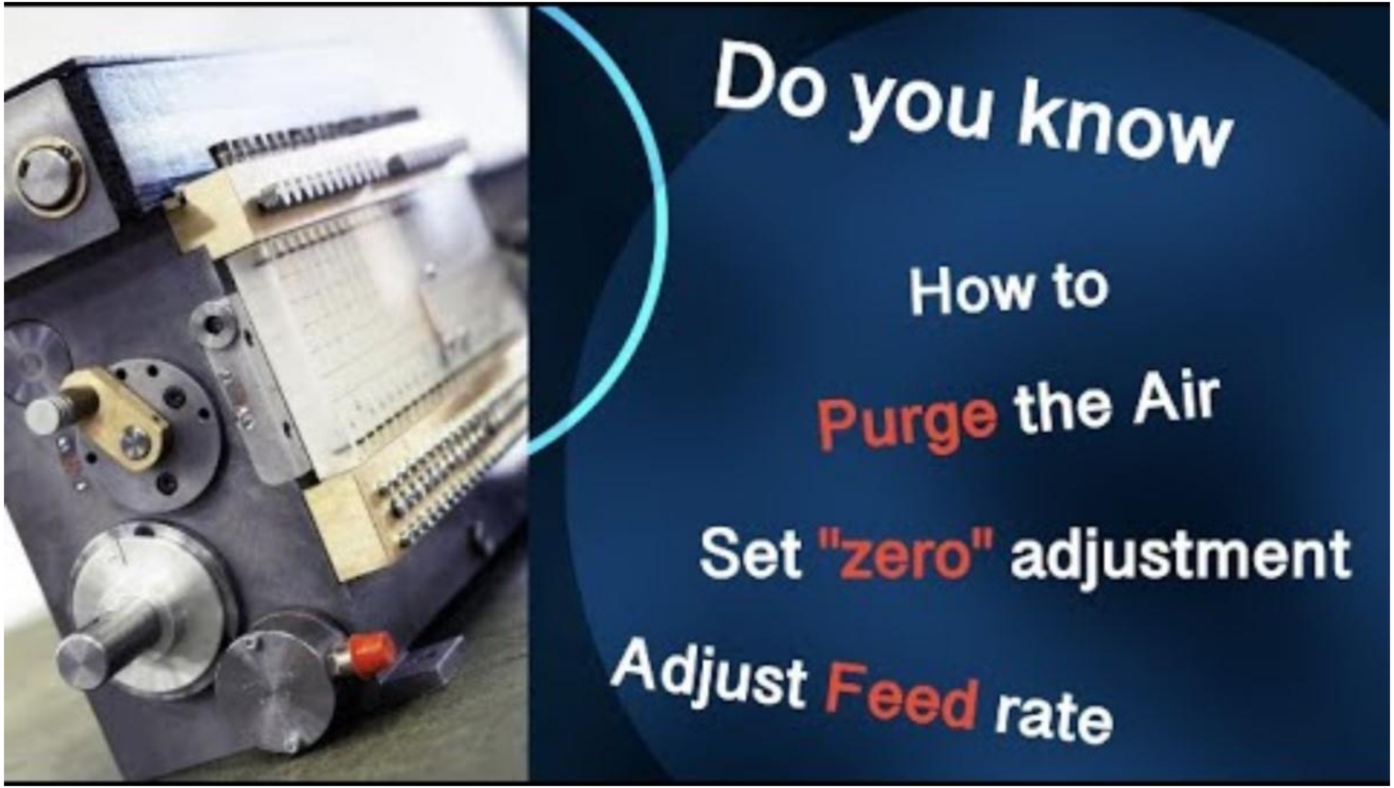
Alpha cylinder lubricator

1:20 – Lubrication oil injection to cylinder liner

Alpha cylinder lubricator



Mechanical cylinder lubricator



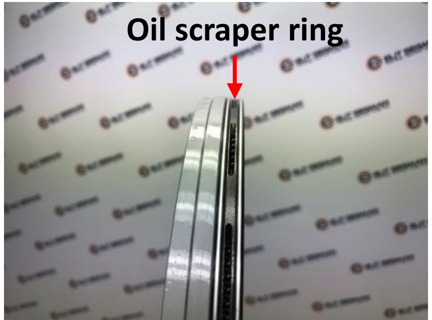
Difference between G/E and M/E

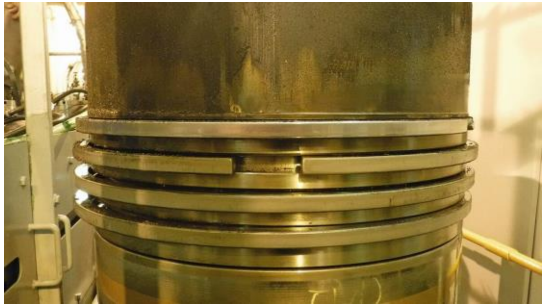
G/E	M/E
Typically, G/E has an oil scraper ring at the very bottom of the piston ring groove and piston rings above it.	Typically, M/Es have identical piston rings fitted into ring grooves.

1 compression ring is being inserted by special ring opener

2 compression rings can be inserted in these ring grooves

1 oil scraper ring was inserted in the ring groove





M/E cylinder liner and piston

Piston and piston rings viewed through the scavenge port of the cylinder liner.

Scavenge port of the cylinder liner as viewed from the top of the piston.

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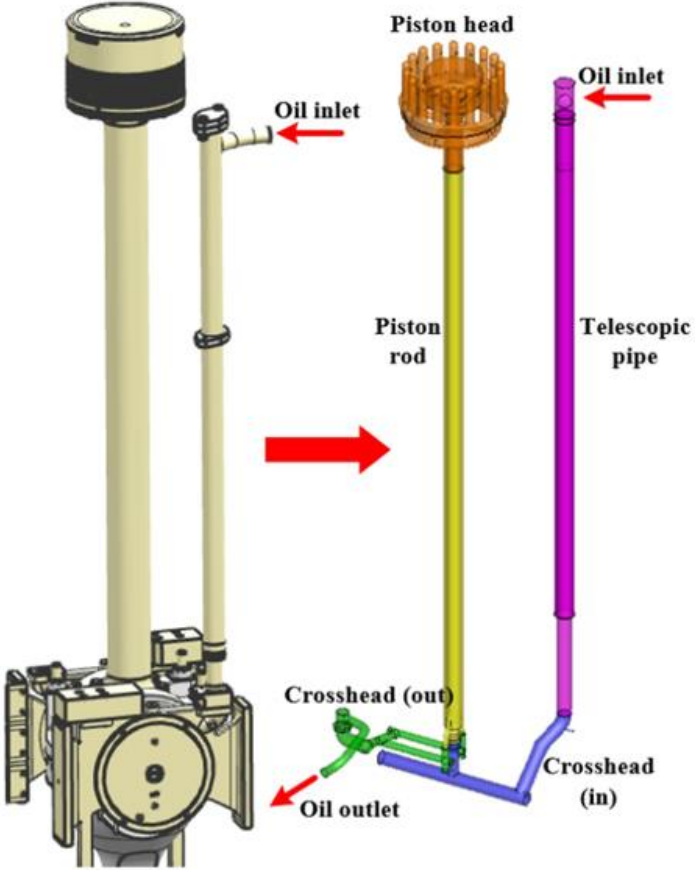
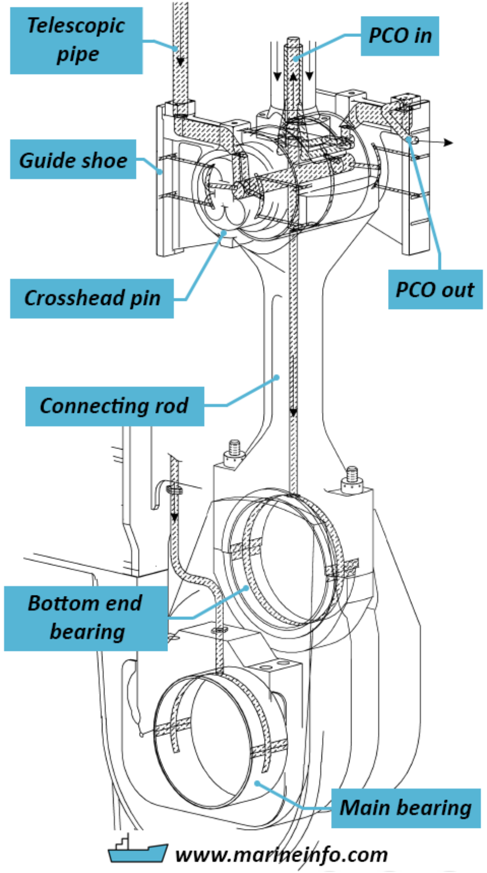
M/E piston cooling





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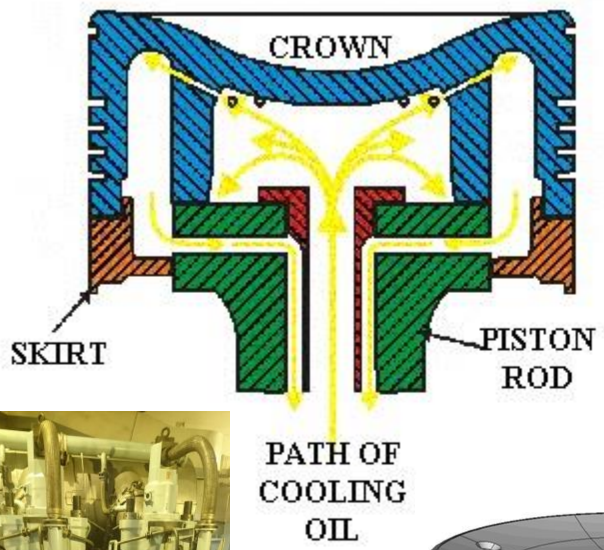
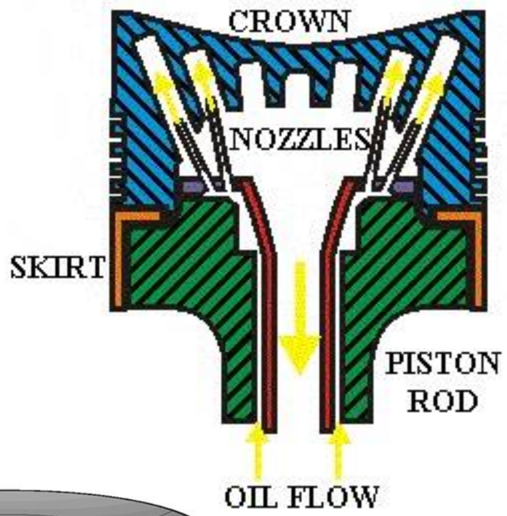
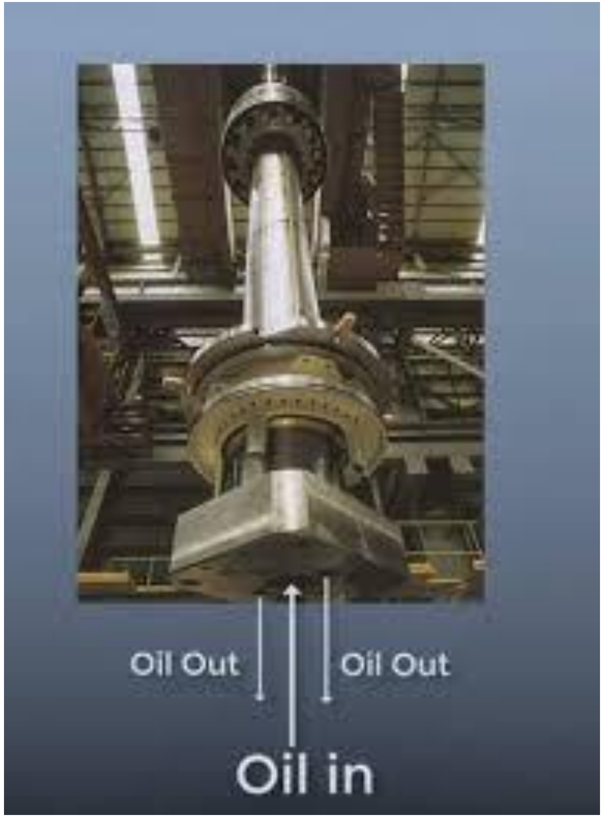
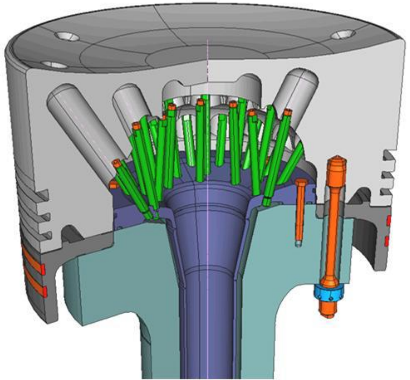
M/E piston cooling

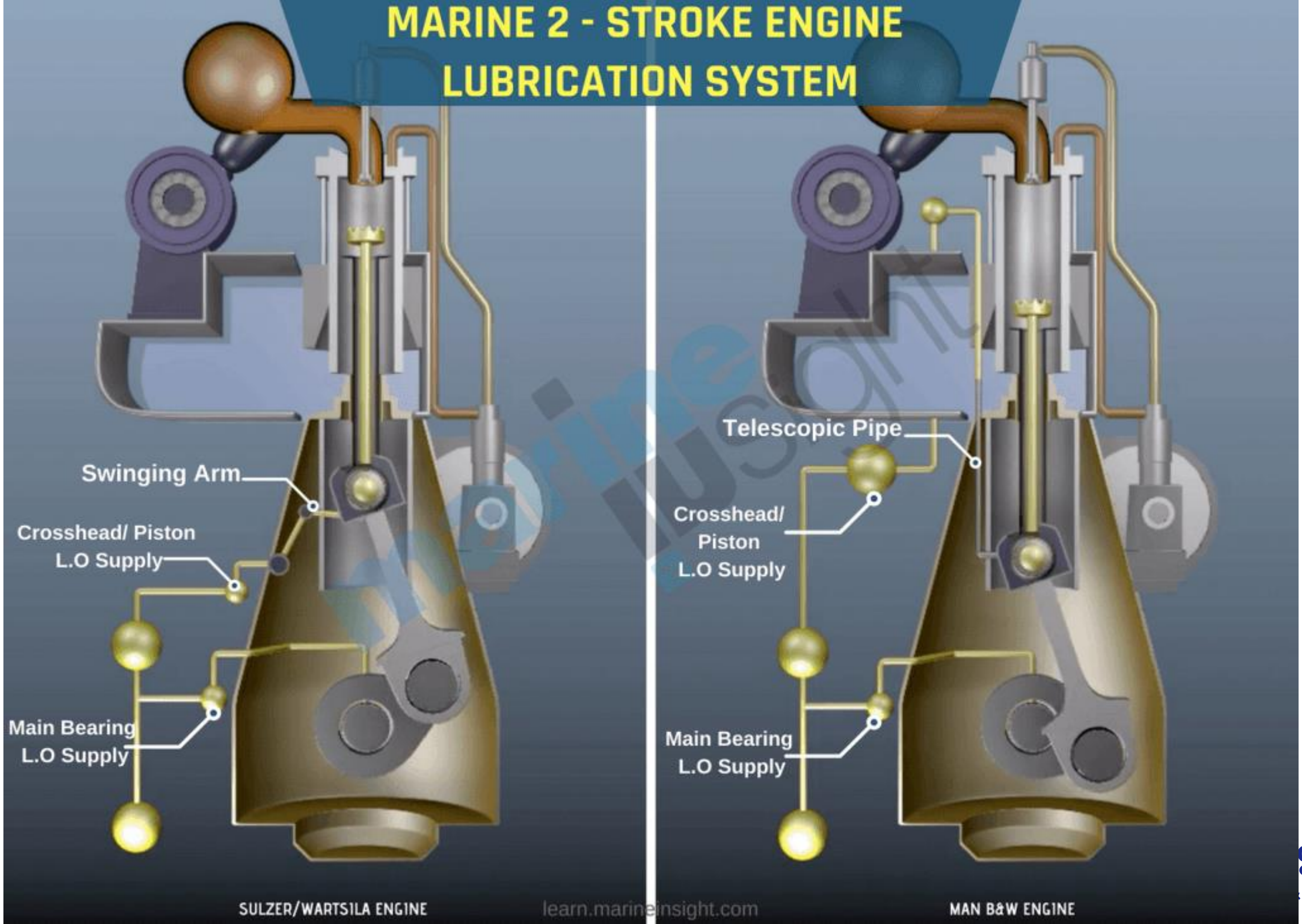


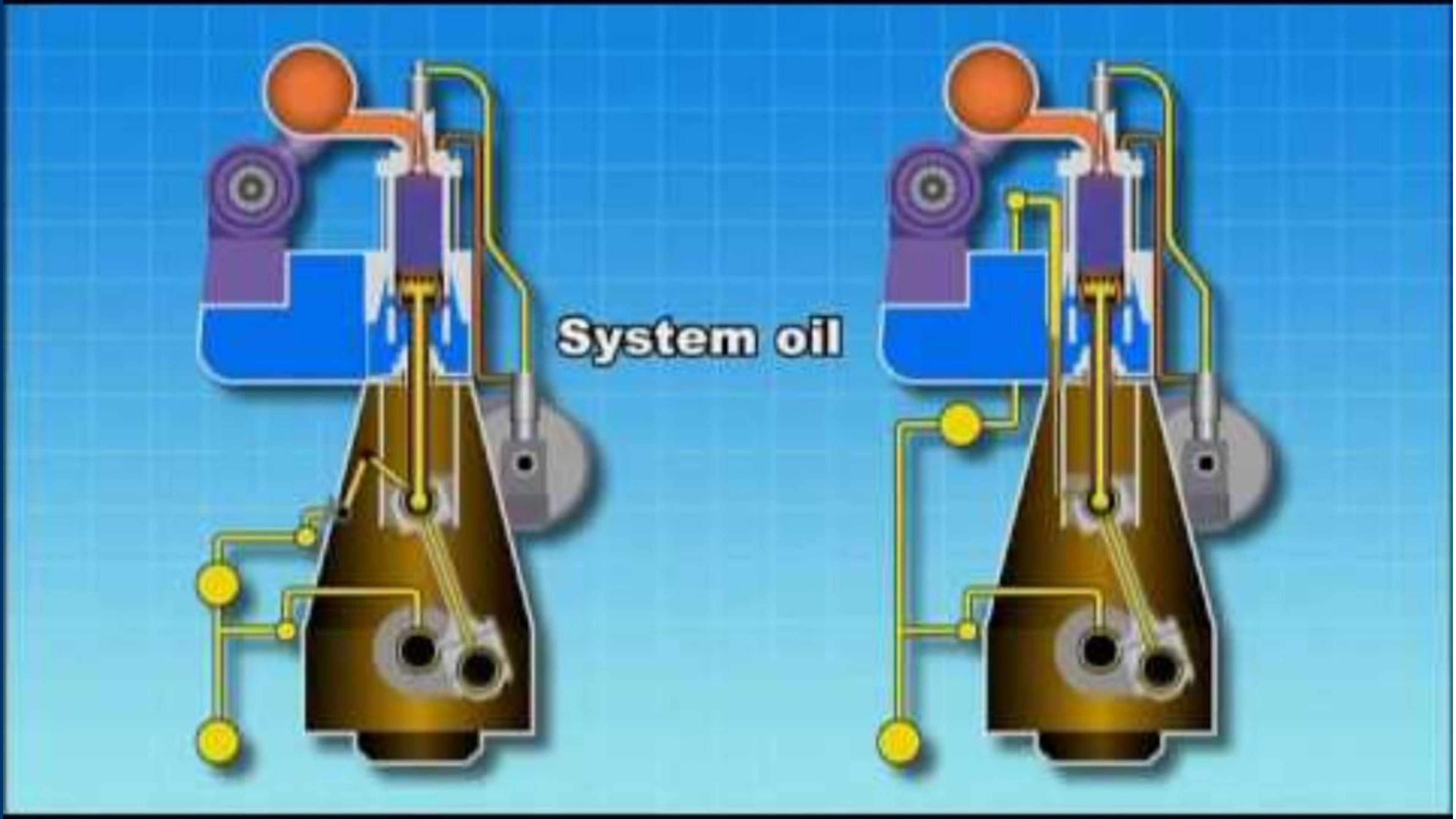






M/E piston cooling


MARINE 2 - STROKE ENGINE
LUBRICATION SYSTEM
Swinging Arm
Crosshead/ Piston
L.O Supply
Main Bearing
L.O Supply
SULZER/WARTSILA ENGINE
learn.marineinsight.com
Telescopic Pipe
Crosshead/
Piston
L.O Supply
Main Bearing
L.O Supply
MAN B&W ENGINE


System oil

Piston rings

Normal and acceptable condition

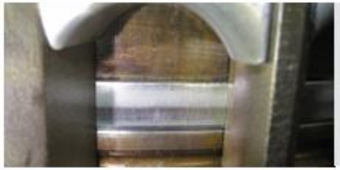
6.1 Normal and acceptable conditions
6.1.1 Piston rings

Ring type: SCP1CC20, A ring
Condition: Normal condition, secondary crack network slightly visible
Acceptance: Normal
Action: No action required

Ring type: SCP1CC20, A ring
Condition: Normal condition, regular secondary crack network
Acceptance: Normal
Action: No action required

Ring type: SCP1CC20, A ring
Condition: Abrasion (3 body abrasion) can be caused by catalyst fines in fuel oil, or foreign hard particles (e.g. sand in intake air).
Acceptance: Acceptable
Action: See WinGD Fuel Guideline - www.wingd.com/en/technology-innovationengine-technology/engine-design/tribology-fuels-lubricants/

Ring type: SCP1RC16, B ring
Condition: RC running-in coating in spotless condition
Acceptance: Acceptable
Action: No action required



Ring type: GTP1CC17
Condition: Abrasive scratches on piston ring running surface by foreign particles
Acceptance: Acceptable
Action: No action required

Ring type: GTP1CC24
Condition: Erosion at GT ring gap by combustion gases
Acceptance: Normal
Action: Monitor Liner Wall Temperature (LWT)

Piston rings

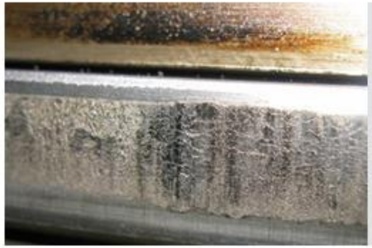
Critical condition

6.3 Action required
6.3.1 Piston rings

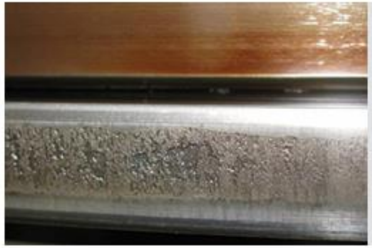
Ring type: GTP1CC20, A-ring
Condition: Local scoring / scuffing of ring gap ends
Acceptance: Critical, to be monitored very closely.
Action: Local dressing up of ring end gaps with emery cloth (80 grain size) through scavenging air ports. If condition deteriorates, then piston ring to be replaced and scorings on liner to be dressed up

Before dressing-up

After dressing-up



Ring type: SCP1CC20
Condition: Active scuffing, CC coating destroyed, active scuffing
Acceptance: Critical condition
Action: Unit to be overhauled. Temporary increase of set cylinder oil feed rate by ~0.2 g/kWh and reduce fuel injection correction factor in the engine control system until unit is overhauled. Consider exhaust gas deviation alarm. If ships schedule permits it, switch off the unit, till unit is overhauled.



Ring type: SCP1CC20, C ring
Condition: Scuffed, CC coating destroyed and cohesive spalling, active scuffing
Acceptance: Critical condition
Action: Unit to be overhauled. Temporary increase of set cylinder oil feed rate by ~0.2 g/kWh and reduce injection correction factor in flexView and reduce fuel injection correction factor in the engine control system until unit is overhauled. Consider exhaust gas deviation alarm. If ships schedule permits it, switch off the unit, till unit is overhauled.

Ring type: SCP2CC20, C ring
Condition: Scuffed with some CC remaining active scuffing
Acceptance: Critical condition
Action: Unit to be overhauled. Temporary increase of set cylinder oil feed rate by ~0.2 g/kWh and reduce fuel injection correction factor in the engine control system until unit is overhauled. Consider exhaust gas deviation alarm. If ships schedule permits it, switch off the unit, till unit is overhauled.

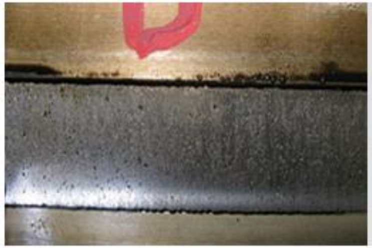
Piston rings

Critical condition

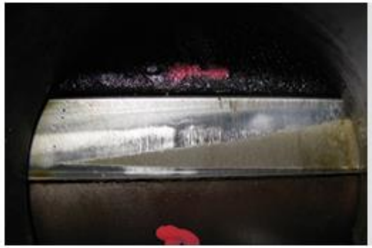
6.3 Action required

6.3.1 Piston rings

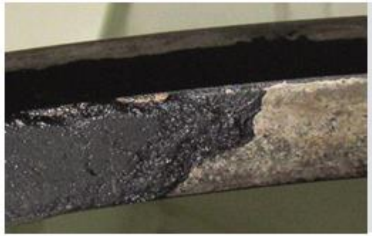
Ring type: GTP1CF24, A ring
Condition: Locally scuffed, CC coating destroyed and spalling, scuffing not active
Acceptance: Critical condition
Action: Unit to be overhauled. Temporary increase of set cylinder oil feed rate by ~0.2 g/kWh and reduce fuel injection correction factor in the engine control system until unit is overhauled. Consider exhaust gas deviation alarm



Ring type: SCP2CC20, D ring
Condition: Partly recovered from scuffing, sharp edges with burrs, coating worn down
Acceptance: Critical condition
Action: Unit to be overhauled. Temporary increase of set cylinder oil feed rate by ~0.2 g/kWh and reduce fuel injection correction factor in the engine control system until unit is overhauled. Consider exhaust gas deviation alarm



Ring type: SCP1CC20, A ring
Condition: CC coating worn down to base material, corrosion on remaining CC coating
Acceptance: Critical condition, further operation of such rings will result in a greater liner wear and increased risk for scuffing
Action: Unit to be overhauled
Remark: See next picture for possible root cause



Ring type: GTP1CF24, A ring
Condition: Excessive deposits on ring inner diameter (backside)
Acceptance: Critical condition, may lead to high ring wear and ring can stick
Action: Unit to be overhauled

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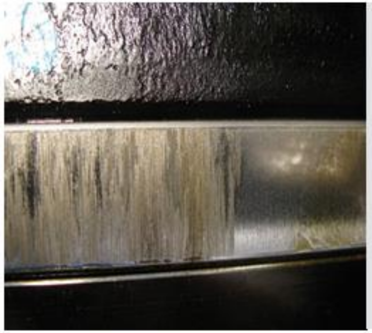
Piston rings

Critical condition

6.3 Action required
6.3.1 Piston rings

Ring type: SCP1CC20, A ring
Condition: Completely worn CC coating
Acceptance: Critical condition, further operation of such rings will result in a greater liner wear and increased risk for scuffing.
Action: Unit to be overhauled



Ring type: SCP1CC20, A ring
Condition: Scoring marks on a worn CC ring
Acceptance: Critical condition
Action: Unit to be overhauled. Temporary increase of set cylinder oil feed rate by ~0.2 g/kWh and reduce fuel injection correction factor in the engine control system until unit is overhauled. Consider exhaust gas deviation alarm.

Ring type: SCP1CC20, A and B ring
Condition: A ring with some corrosion, B ring lost tension, with excessive deposits, cylinder oil feed rate, BN and fuel oil sulphur content not matching
Acceptance: Critical condition
Action: Unit to be overhauled. Check correct cylinder oil feed rate setting. Check residual BN in piston underside drain oil.

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Piston rings

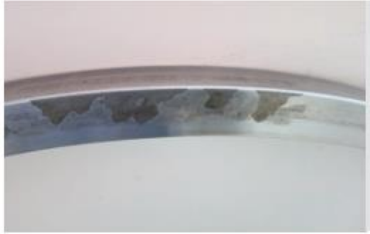
Critical condition

6.3 Action required
6.3.1 Piston rings

Ring type: SCP1CC20, A ring
Condition: Ring collapsed, with excessive deposit
Acceptance: Critical condition
Action: Unit to be overhauled. Switch-off unit

Ring type: SCP1CC20, A ring
Condition: Ring broken
Acceptance: Critical condition
Action: Unit to be overhauled. Switch-off unit

Ring type: SCP1CC16, A ring
Condition: A ring with excessive spalling across full ring height
Acceptance: Critical condition
Action: Unit to be overhauled. Temporary increase of set cylinder oil feed rate by ~0.2 g/kWh and reduce fuel injection correction factor in the engine control system until unit is overhauled. Consider exhaust gas deviation alarm



Ring type: GTP1CF24, A ring
Condition: Bottom face Cr coating spalling
Acceptance: Critical condition
Action: Piston ring to be replaced

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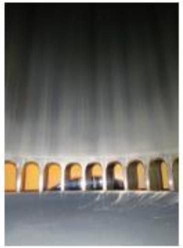
Cylinder liner

Normal and acceptable condition

6.1 Normal and acceptable conditions
6.1.2 Liner


Condition: Spotless condition of liner, honing marks clearly visible
Acceptance: Normal
Action: No action required




Condition: Good liner condition, smooth and homogenous liner appearance, honing marks not visible anymore
Acceptance: Normal
Action: No action required

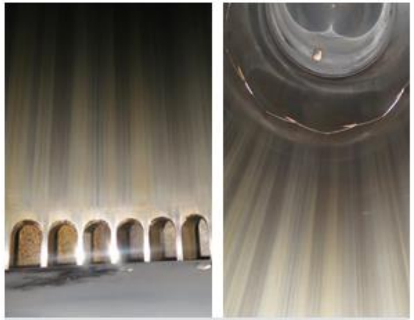

Condition: Cylinder liner with minor cold corrosion (milky spots)
Acceptance: Normal
Action: No action required

Condition: Single scratches on liner surface
Acceptance: Normal
Action: No action required

Cylinder liner

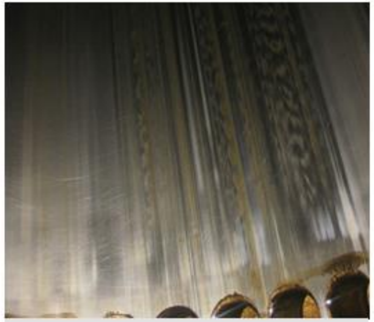
Critical condition

6.3 Action required
6.3.2 Liner



Condition: Cylinder liner with scoring marks
Acceptance: Critical condition
Action: Depending on piston ring condition unit should be overhauled and liner re-honed. If piston rings are in spotless condition, the unit can be kept in operation.

Condition: Water ingress caused rusty liner surface.
Acceptance: Critical condition
Action: Source of water leak to be found and rectified.



Condition: Liner with local scoring and scuffing, marks
Acceptance: Critical condition
Action: Depending on piston ring condition unit should be overhauled and liner dressed-up according to the procedure described in chapter 6.5 or re-honed. If piston rings are in spotless condition, the unit can be kept in operation.

Condition: Scuffing marks on liner
Acceptance: Critical condition
Action: Unit to be overhauled. Local scoring and scuffing marks are to be dressed up according to the procedure described in chapter 6.5 or re-honed.

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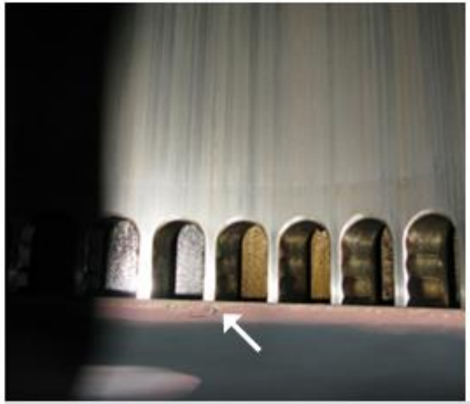
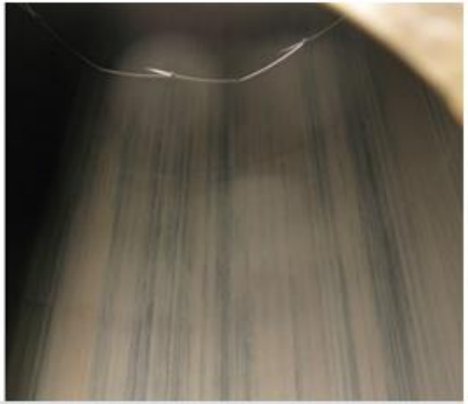
Cylinder liner

Critical condition

6.3 Action required
6.3.2 Liner



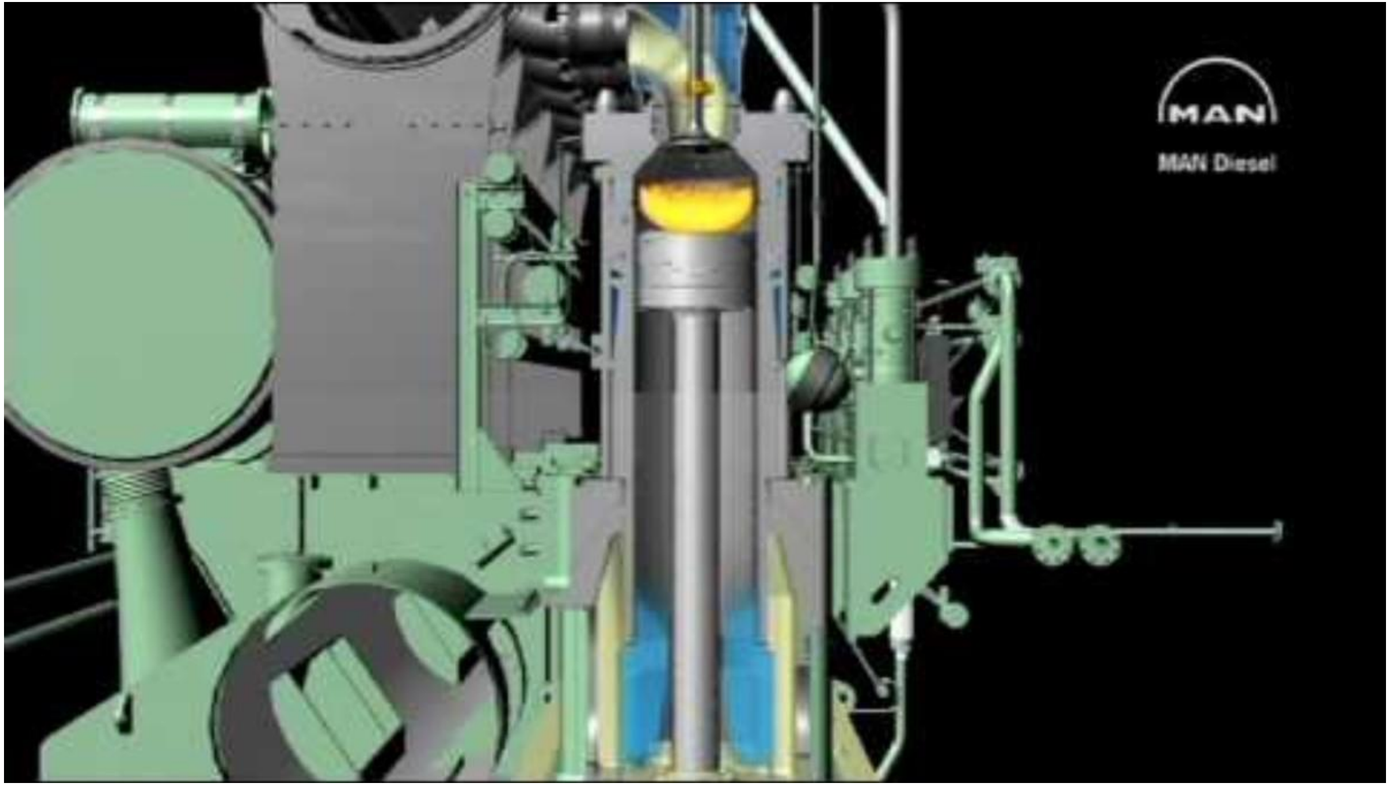


Condition: Totally scuffed unit with dull appearance. Note the reddish spots on the piston crown top, which are oxidised iron from the liner

Acceptance: Critical condition

Action: Such a liner has to be replaced as soon as possible

M/E working principle



References

1) https://www.behance.net/gallery/31936141/EXXONMOBIL-FPSO-Vessel/modules/203400195

2) https://www.youtube.com/watch?v=aoZTPjAkTek

3) https://www.alibaba.com/product-detail/Ship-Diesel-Engine-Valves-EXHAUST-VALVES_60657967593.html

4) https://www.savree.com/en/encyclopedia/marine-two-stroke-engine-exhaust-valve

5) https://muckymariners.com/app/meo-class-4-subjects/function-4b-meol/engines/9-exhaust-valve/exhaust-valve-3-5-main-engine-exhaust-valve-overhaul/

6) https://www.tradekorea.com/product/detail/P349737/Exhaust-valve.html

7) https://www.youtube.com/watch?v=T0H3hPGbyhU

8) https://www.dieselduck.info/machine/01%20prime%20movers/2010%20-%20MAN%20Project%20Guide%206S90ME.pdf

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References

9) https://en.ppt-online.org/322268

10) https://www.savree.com/en/encyclopedia/air-start-valve

11) https://www.man-es.com/docs/default-source/service-letters/sl2018-668.pdf?sfvrsn=5c517c65_4

12) https://www.scribd.com/document/486011736/90302-pdf

13) https://shipfever.com/starting-air-distributor/

14) https://www.youtube.com/watch?v=hoNMq-e06zs

15) https://chirp.co.uk/report/removal-of-heat-protection-from-machinery/

16) https://www.youtube.com/watch?app=desktop&v=6QrTpx5J9FU

References

17) https://wooriengine.com/?page_id=21&mod=document&uid=34

18) https://www.youtube.com/watch?v=5LvGC-3t5y8

19) https://mirmarine.net/component/content/article/391-exhaust-valve-procedures-for-inspection-and-calibration?layout=blog&Itemid=&catid=148

20) https://chiefengineerlog.com/2024/04/01/the-ships-main-engine-scavenge-receiver/

21) https://marinersgalaxy.com/main-engine-exhaust-valve-parts-working-overhaul-procedure/

22) https://www.researchgate.net/publication/340032791_DVIGUNI_WinGD_TIPU_RT-flexX_Z_ELEKTRONNIM_UPRAVLINNAM_ELECTRONICALLY_CONTROLLED_WinGD_RT-flexX_ENGINES

23) http://www.dongjin-ms.com/doc.php?idx=57

24) https://gallois.be/ggmagazine_2006/gg_01_01_2006_7.pdf

References

25) https://ciss-group.com/sertificate/Presentation%20CISS%20Group%20-%20repair.pdf

26) https://journals.sagepub.com/doi/abs/10.1177/1468087419872113

27) https://www.youtube.com/watch?v=6YDaK5Hbj1M

28) https://www.marineinsight.com/main-engine/intelligent-cylinder-lubrication-for-modern-marine-engines-part-1/

29) https://www.elitsegman.com.tr/hyundai-himsen/hyundai-himsen-h2533-piston-rings-set/

30) https://www.rivieramm.com/news-content-hub/news-content-hub/new-cylinder-lube-oils-more-than-just-bn-index-56035

31) https://www.almship.com/shop/s60mc-piston-rings-man-b-w-s60mc-piston-rings-3621#attr=

32) https://wingd.com/media/odqa15kl/wingd-guide-for-judging-condition-of-piston-running-components_v6.pdf

References

33) https://wingd.com/media/odqa15kl/wingd-guide-for-judging-condition-of-piston-running-components_v6.pdf

34) https://www.youtube.com/watch?v=OpqgIzw6xa4

35) https://www.youtube.com/shorts/vHi-bOe47ek

36) https://www.youtube.com/watch?v=XfZgyYXAd0g

37) https://www.youtube.com/watch?v=R9N12cUTEDg

38) https://www.vitatr.com/services/overhaul-and-adjustment-of-cylinder-lubrication-systems-2

39) https://savree.com/en/encyclopedia/marine-2-stroke-engine-stuffing-box

40) https://m.blog.naver.com/newrains/222710492197

References

41) https://www.youtube.com/shorts/JgDrN7dPAfA

42) https://www.youtube.com/shorts/-cpmr_04I30

43) https://www.youtube.com/watch?v=zH7V1NbNlPM

44) https://www.team-bhp.com/forum/commercial-vehicles/181906-explained-how-marine-diesel-engines-work.html

45) https://www.youtube.com/live/Ed2xKhMCgMc?app=desktop

46) https://etoinsights.com/2024/08/09/a-brief-overview-of-me-engine-control-system-8/

47) https://www.youtube.com/watch?v=Cz5LPIStbV8

48) https://www.marineinsight.com/tech/ships-main-engine-lubrication-system-explained/

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References

49) https://www.sciencedirect.com/science/article/pii/S1359431119345259#f0005

50) https://www.youtube.com/watch?v=c-IaMlEF3bk

51) https://marineengineering4all.blogspot.com/2017/05/marine-diesel-engines-2-stroke.html

52) https://marinerspointpro.com/piston-cooling-in-2-stroke-marine-diesel-engine/

53) https://marineinfo.com/main-engine-lubricating-oil-system/

54) https://www2.kaiyodai.ac.jp/~takagi/ime/0604_Piston_Cooling.pdf

55) https://www.reddit.com/r/Skookum/comments/d5fkuv/so_i_heard_you_like_big_pistons/

56) https://link.springer.com/article/10.1007/s42835-020-00375-z

57) https://www.youtube.com/watch?v=pxK97xxr_AI

References

58) https://www.facebook.com/malaayjoseph/posts/basic-informationauxiliary-blowera-main-engine-auxillary-blower-is-used-to-provi/663958612865976/

59) https://www.facebook.com/masterengineerfix/posts/auxiliary-blowers-boosting-efficiency-in-large-2-stroke-marine-diesel-engines-%EF%B8%8Fh/1121211436672417/

60) https://www.facebook.com/MarinoMakinistangMalupit/posts/what-are-auxiliary-blowers-in-2-stroke-main-engines-during-start-up-if-there-is-/119869252777375/

61) https://accelleron.com/charge-magazine/turbocharger-maintenance-tips

62) https://www.turbo.co.jp/en/service/basics.html

63) https://www.marineinsight.com/tech/10-points-efficient-turbocharger-operation/

64) https://www.youtube.com/watch?v=QOtN7yFXP8Y

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